

Research Article

Photovoltaic Solar Technologies: Solution to Affordable, Sustainable, and Reliable Energy Access for All in Rwanda

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Despite remarkable economic growth and development in recent decades, Rwanda has been still facing energy crises and challenges. Although the country has considerable energy assets, less than 10% is utilized for its local electricity needs. Currently, national installed generation capacity is estimated at 221 MW, for a population around 12 million, and electricity access is estimated at 51% (37% grid and 14% off-grid networks). About half the population is without electricity access while the grid-connected users face high electricity tariffs and frequent power outages (blackouts). The national grid itself is also experiencing high losses. This paper used the HOMER software for modeling the optimal, sustainable, reliable, and affordable photovoltaic solar technologies as energy solutions for all (off-grid and on-grid users) in Rwanda. The selection and recommendation of a suitable photovoltaic (PV) solar technology depend on its annual electricity production capacity, electrical load, renewable energy penetration percentage, economic viability, feasibility, affordability, carbon footprint, and greenhouse gas emission level for climate change considerations towards a clean and greener future. The results show that the least cost of energy (LCOE) for electricity production by each of the solar PV systems with storage, PV-grid-connected household, and PV-grid connection with storage was 67.5%, 56.8%, and 33.9%, respectively, lower than the normal electricity tariff in Rwanda. The PV systems with storage proposed in this paper could be effective in increasing national energy resource exploitation, providing affordable and reliable energy access to all citizens.

1. Introduction

Energy is among the main factors and pillars to modern society whose modern day generation is dominated by industrialization and technology improvement [1]. Meanwhile, some countries are still facing problems related to lack of energy access and challenges from unreliable (grids characterized by frequent blackouts and power outages day to day) electrical grids. Insufficient electricity access and unreliable grids limit business activities, work-life balance, and provision of social amenities [2].

Although over 600 million people do not access electricity in sub-Saharan Africa, more millions of people are connected to mostly epileptic electricity grids unable to provide their everyday energy requirements [3, 4]. While 20% electricity access rate is prevalent [5, 6], only a third can access contemporary energy facilities [7, 8]. Further, the power outage frequency ranges between daily and once in four days [9]. These occurrences have negatively impacted economic productivity while increasing costs of doing business in the region [2, 3, 10, 11].

Rwanda, a country in sub-Saharan Africa, is among candidate countries facing lack of energy access to all citizens

and reliable grid. Currently, Rwanda has only 221 megawatts of installed electricity generation with 51% electricity access (37% national grid and 14% mainly solar) [12]. The national power network presents high percentage losses, sometimes going beyond 30.0% [13]. Small and ageing electrical grid infrastructures, rapidly increasing population, and widening supply-demand divide require the provision of renewable energy resources to supply the shortfall. With such imbalance of the population's energy demand and available capacity, the government of Rwanda sets to achieve 512 MW installed power generation capacity in 2023/2024 [12, 14].

Also, there are lots of local natural energy resources which have not yet been well exploited at maximum [12]. In the solar energy sector, Rwanda is located about 2 degrees south of the equator making it excellent for solar energy development, with 8.5 MW grid-connected and operational solar energy in the energy generation mix. The country's insolation is between 4.5 and 5 kWh per m² per day with 5 peak sun hours [15], which are favorable for solar energy. Further, private sector participation is required to provide solar lighting solutions to remote areas in the country.

Thus far, the independent power producers (IPPs) have extended electricity access to more than 258414 households using solar energy nationwide. Rwanda is temperate with a network of rivers, whose climate is greatly influenced by its altitude. The rainfall figures hover between 900 mm in the east and 1600 mm in the west. Therefore, hydropower continues to be a good natural electricity resource for Rwanda [12, 16].

The facts of population growth coupled with economic and modern technology desires in Rwanda need urgent, reliable, and sustainable electrification strategies and technologies with high generation rates as many researchers have stated. Hence, more and more electricity generation capacity will be required by this country so that it can meet its growing electricity energy demand and targets. The electricity generation capacity of Rwanda is lower than its demand, especially because around 50% of the population is without electricity and energy access.

Besides the half lacking electricity access, the power network has high percentage losses beyond 30.0% and the electricity tariff is the most expensive in the East African Community (EAC) [12, 17]. Also, households accessing electricity from the grid face challenges related to unreliable supply characterized by frequent power outages and electricity blackouts which usually result in damaging some very costly household equipment. Sometimes, it can result in loss of human lives due to sudden return of electricity during the blackouts. With these challenges and many more, the government has set out to resolve some of these problems for the citizenry.

To fulfill and achieve such targets and objectives, there must be an increase in exploitation of local natural energy resources [12, 17]. Rwanda's geography is suitable for solar energy, but this sector has not been fully exploited. Once this country can exploit its solar energy potential at maximum, photovoltaic solar energy technologies can solve a lot of the problems of unreliable grid, high electricity tariffs, and lack

of electricity access, and then, it can contribute to the national economic growth, development, and worldwide environmental sustainability.

The paper investigated, analyzed, and described the solar energy potential in Rwanda and how different photovoltaic solar energy technologies can help the government in meeting and achieving its energy plans, targets, and objectives. The highlights of objectives are to evaluate, investigate, and analyze the photovoltaic solar energy technologies (abundant natural and local energy resource in Rwanda which has not yet been well exploited) with respect to the government energy targets, objectives, and plans that establish viable solutions for both off-grid and on-grid residents. Not only these but also the economic viability and feasibility of exploiting photovoltaic solar energy technologies within the country's context of achieving its targets and objectives on an electricity generation capacity increase and electricity demand were covered.

If solar energy is well exploited through different photovoltaic technologies, this can lead the country to different solutions, like ensuring affordability, reliability, and continuous electricity access to all citizens nationwide, increasing local natural energy resource exploitation, and fostering and optimizing the power capacity and environment conservation, self-reliance, and stability of electricity supply and generation.

However, the government is in search of innovative technologies for local natural renewable energy technology exploitation, and such solar energy technologies were developed in this paper. Therefore, for off-grid users, the first time HOMER software is used to identify and determine the most optimal PV technologies for off-grid areas that can power both AC and DC loads in Rwanda. Further research into the PV off-grid sector for Rwanda can consider battery sharing, PV sharing for nearby households, and much socio-techno-economic conditions for the purpose of PV technology-wide dissemination in many places and to all people in need. For grid-connected users, further research can consider modeling improved technologies by including many household factors and the impact of high PV penetration on the national grid and assessing the willingness of Rwandan citizens to accept new PV technologies towards increasing self-produced (local) electricity generation capacity.

The paper was divided into Introduction; Materials, Methods, and Literature; Solar Energy in Rwanda; Modeling and Optimization of Off-Grid PV Systems; Modeling and Optimization of Grid-Connected PV Systems; and Conclusion.

2. Materials, Methods, and Literature

Site visits were done in Western Province of Rwanda (Rutsiro District, Rutsiro, Rwanda (1°56.3'S, 29°19.5'E), and Karongi District, Kibuye, Rwanda (2°3.5'S, 29°21.0'E)) to different households and habitats in the community. A daily electrical load for an individual household and a daily electrical load for 100 households in one community were estimated

TABLE 1: Residential house electrical equipment and daily load outline.

No.	Equipment	Quantity	Power rating (W)	Hours of use/day	Daily energy usage
1	Lights	8	15.0	12.0	1440.0
2	Lights	14	15.0	4.0	840.0
3	Computers (laptop)	3	25.0	5.0	375.0
4	Washing machine	1	245.0	5.0	1225.0
5	Shaver	1	15.0	0.6	9.0
6	Iron	1	1100.0	0.6	660.0
7	TV	2	125.0	6.0	1500.0
8	Cell phone	6	2.5	7.0	105.0
9	Satellite receiver	1	25.0	24.0	600.0
10	Refrigerator	1	150.0	3.0	450.0
Annual load = 2630.0 kWh/year					

(Tables 1 and 2). The data (stream flow rate) of the Mukungu River located in Karongi District, Western Province of Rwanda, was collected, processed, and used in the analysis of hybrid technology of combining solar photovoltaic system and hydro. Data from EICV 5 (Integrated Household Living Conditions Survey, 2017) through the Statistical Package for the Social Sciences (SPSS version 24) on living conditions in Rwanda (number of power outages per week) were processed and analyzed to assess the usage possibility of solar energy technologies once adopted at high penetration. The study in this paper also analyzed the performance of the Rwamagana 8.5 MW solar power plant. The Mann-Kendall Test was used to test for the trend of energy fed into the grid by this solar power plant.

Trend analysis assesses whether the values for a series of observations over time are increasing, decreasing, or constant. Mann-Kendall monotonic trend analysis does not depend on the estimate of the trend itself, but on relative ranking of the data [18, 19]. It is implemented either as the classical or seasonal Mann-Kendall Test [20]. The existing hypothesis (H_0) derives from identically independent distributed (iid) data of a population, while the researchers' hypothesis (H_a) could be increasing or decreasing monotonically [21]. The Mann-Kendall Test score is evaluated from Equations (1)–(6) [22]:

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(X_j - X_k), \quad (1)$$

because

$$\text{sgn}(x) = \begin{cases} 1, & \text{if } x > 0, \\ 0, & \text{if } x = 0, \\ -1, & \text{if } x < 0. \end{cases} \quad (2)$$

The mean of S is $E[S] = 0$, and

$$\text{variance}(\sigma^2) = \frac{n(n-1)(2n+5) - \sum_{j=1}^p t_j(t_j-1, 2t_j+5)}{18}, \quad (3)$$

where p is tied group number and t_j is the data point number in the j^{th} tied group. The S score is nearly normally distributed using the Z -transformation:

$$Z = \begin{cases} \frac{s-1}{\sigma}, & \text{if } S > 0, \\ 0, & \text{if } S = 0, \\ \frac{s+1}{\sigma}, & \text{if } S < 0. \end{cases} \quad (4)$$

The S score is connected to Kendall's T because

$$T = \frac{S}{D}, \quad (5)$$

where

$$D = \sqrt{\left[\frac{1}{2}n(n-1) - \frac{1}{2}\sum_{j=1}^p t_j(t_j-1) \right] \left[\frac{1}{2}n(n-1) - \frac{1}{2}\sum_{j=1}^q v_j(v_j-1) \right]}. \quad (6)$$

The Hybrid Optimization of Multiple Energy Resources (HOMER) Pro software was used to model, analyze, and optimize possible photovoltaic solar technologies as a solution to sustainable and reliable energy access for all in Rwanda. It simulates and optimizes both conventional and renewable electric power system generation sources. It helps in designing the most cost-reflective systems by analyzing thousands of power systems in a very short time [23].

The HOMER software is a leading world power system analysis tool for the design, modeling, and optimization of microgrid technologies [24, 25]. It can model almost every form of energy technologies used in conventional and renewable energy resources and technologies. It can also model and optimize energy systems for small and medium firms provided relevant input electricity consumption load outlines and other parameters are available and accessible [25].

Although small and medium grids can be modeled and optimized for small and medium firms in Rwanda, they must comply with government regulations. Therefore, in hybrid renewable and nonrenewable energy technologies, the larger proportion of between 50 and 70% of the aggregate electricity production must be from renewable resources. If it is a hybrid solar-diesel minigrid, the larger electricity production of between 50 and 70% must be solar. Others include creating awareness and education of energy business activities between potential consumers, identifying mostly used electrical equipment, enabling firms to access capital for equipment acquisition, training on efficient energy uses to reduce consumption costs, diversification, and sustainable growth.

TABLE 2: Community electrical loads and their daily load profile.

No.	Equipment in 100 households	No. in use	Power consumption (W)	Total power consumption (W)	Hours of use/day (hr)	Watt-hours/day
1	Lamps	744	10.0	7440	5	37200
2	Cell phones	280	10.0	2800	8	22400
3	Ceiling fan	0	75.0	0	0	0
4	Radio	94	20.0	1880	5	9400
5	TV	50	127.6	6380	5	31900
6	Computer	3	100.0	300	5	1500
7	Refrigerator	4	500.0	2000	24	48000
8	Iron	15	1000.0	15000	1	15000
Daily total energy consumption in 100 households					165.44 kWh/day	
Average annual energy consumption in 100 households					60385.60 kWh/year	

In order to develop a minigrid project in Rwanda (institutions or companies), there may be documents or information required by documents or information may be required including but not limited to licencing, permits, sustainable operations, cost recovery plans, tariff regulations, financial support programmes, quality standards, status of connecting the minigrid to the national main grid in case the minigrid project is located not far of the national main grid, or what will happen upon arrival of the main national power grid in the future (this is in the case when the minigrid is constructed far of the national main grid). The minigrid developer must also provide for installation of prepaid electricity meters to the ultimate consumers and automate the systems so that consumers can buy electricity units from retail shops, offices, and other outlets.

The combination of levelized energy cost, benefits of many generation assets, fuels used in production, and the byproducts of generation enables same energy per unit cost comparisons. Thus, the energy cost method has the disadvantages of sometimes not being able to produce energy as the byproducts, inability to obtain exact equivalencies between other product costs and electricity costs, and the problems of estimating electricity costs below their actual value [26].

These multiplied energy problems are nonlinear so that we can place economic flexibility limits on electricity generation. The algorithm is developed using electricity capacity output unit cost (c_e), fixed operating output unit cost (f_e), and variable output unit cost (v_e). The PV electricity generation cost becomes [27]

$$\text{CoPV}_e = c_e + f_e + v_e. \quad (7)$$

The per unit kilowatt (kW) capacity cost equals

$$c_e (\$/kW_e) = \frac{P_e}{n \cdot \text{CF} \cdot \sum_{j=1}^Y a_j \delta^j}, \quad (8)$$

where P_e is the unit capacity price for a kW an hour. Capacity cost comprises costs for preliminary designs, bills of quanti-

ties, land and buildings, imports, valuation, and other project management and consultancy services. The PV system is serviceable for Y years and produces $n \cdot a_j \cdot \text{CF}$ kW in year j . Further, n equals 8760 hours a year, a_j is the loss factor, CF is the facility factor, and δ is the reduction factor. Also, the reduction factor increases with time. It produces interperiod factors for interest rate computation. It eases audit processes when modeling with discounted cash flow. It also replaces either the net present value (NPV) or internal rate of returns (IRR). Additionally, ξ is the reduction rate, which changes once-off costs into annualized costs. The reduction factor is calculated as [15, 26, 27]

$$\text{reduction factor} = \frac{1}{1 \times (1 + \text{reduction rate})^{\text{period number}}}. \quad (9)$$

The simplified reduction factor in energy projects at a breakeven point becomes

$$\delta = \left(\frac{1}{1 + \xi} \right)^2. \quad (10)$$

Also, the operational fixed costs fluctuate by the year while the mean operational fixed costs per kilowatt in any year equal $f_{ej} (\$/kW_e)$. Further, these operational fixed costs are not functions of the real electricity generated by the PV systems:

$$f_e = \frac{\sum_{j=1}^Y f_{ej} \delta^j}{n \cdot \text{CF} \sum_{j=1}^Y a_j \delta^j}. \quad (11)$$

If $v_{ej}(t) (\$/kW_e)$ is the time-changeable cost of generating 1 kW by year j and $v_{ej} (\$/kW_e)$ is the mean annual changeable cost, which equals

$$v_{ej} = \frac{1}{n} \int_0^n v_{ej}(t) dt. \quad (12)$$

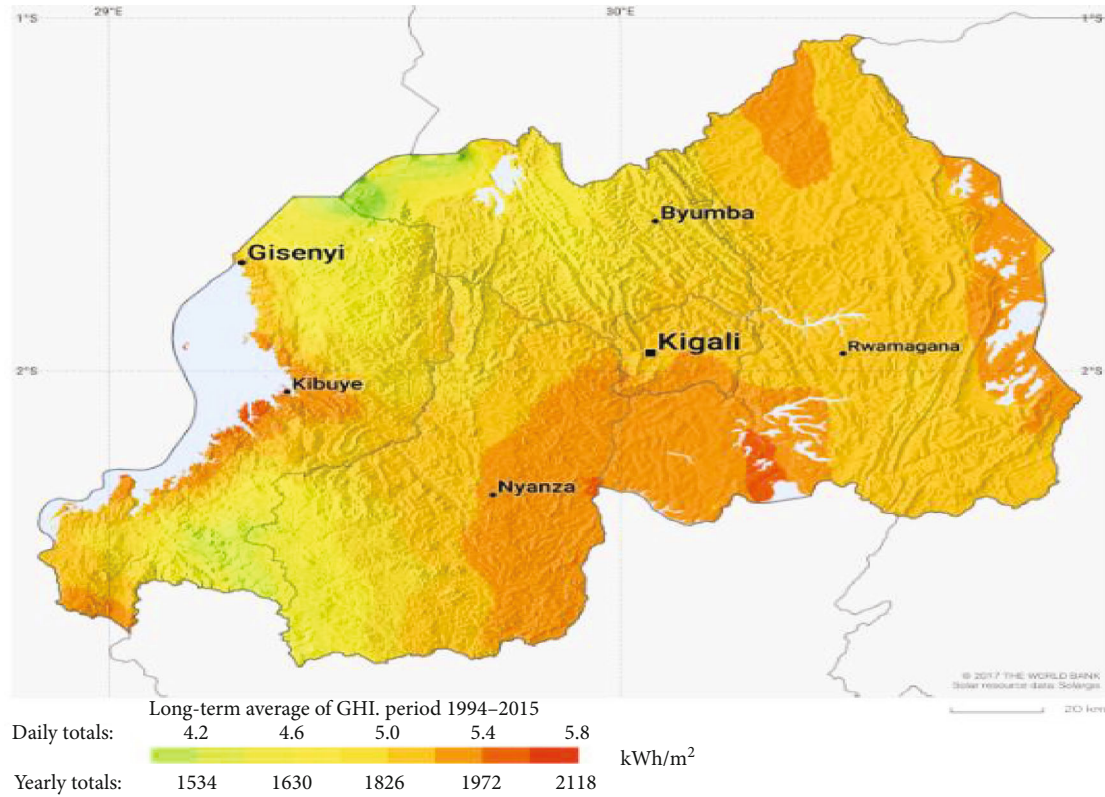


FIGURE 1: Global horizontal irradiation for Rwanda (1534–2018) [28].

During the project lifespan, the mean time-changeable cost for unit kilowatt equals

$$v_e = \frac{\sum_{j=1}^Y v_{ej} \cdot n \cdot CF \cdot a_j \delta^j}{n \cdot CF \sum_{j=1}^Y a_j \delta^j} = \frac{\sum_{j=1}^Y v_{ej} a_j \delta^j}{\sum_{j=1}^Y a_j \delta^j}. \quad (13)$$

Correspondingly, $P_{ej}(t)$ is the time-sustained electricity price in year j , P_{ej} is the mean annual electricity price by year j , and P_e is mean time electricity price in year j . Therefore, the cost-reflective COE for the PV generation system is cost-efficient whenever

$$P_e > \text{CoPV}_e. \quad (14)$$

Hence, cost efficiency of the PV system enables the required sustainably high net present worth, which is the stamp of authenticity of breakeven analyses.

3. Solar Energy in Rwanda

3.1. Brief Information about Solar Energy in Rwanda. Rwanda's solar insolation is 5 kWh/m²/day and daily 5 peak sun hours. Such radiations and other climatic weather conditions in Rwanda prove that solar energy would significantly contribute to national electricity generation once well exploited. Also, data on the global horizontal irradiation (GHI) map

of Rwanda (1534–2018) classify this country as good for solar energy (Figure 1) [28].

3.2. Analysis of the 8.5 MW Rwamagana Solar Power Source. The analysis of 8.5 MW Rwamagana solar power source indicates that it fed 4175459.89 kWh to the national power grid within three years (January 2015 to December 2017). The minimum feed-in was 988730 kWh and the maximum was 1423200 kWh during January 2016 and August 2016, respectively. An annual average of 1391808.63 kWh was fed to the national grid during the 3 years. Besides synchronization, the power factor of an inverter of a grid-connected solar power system must be equal or very close to unity [29] because it was designed to generate real power at a unity power factor [30]. For the Rwamagana gigawatt solar power plant, the power factor data as it was remarked during the site visit were equal and close to a unity power factor. The frequency levels were close to 50 Hz [13] which is the standard frequency for the national power grid in Rwanda. The analyses of energy feed-in in the national power grid (kWh), frequency (Hz), and power factor show that the Rwamagana power plant is in a good operational mode, but further analysis of factors may be considered for future improved and expanded research on operational status of this solar power plant (Figures 2–6 are the authors' calculation and plotting based on the data collected during the site visits at Rwamagana 8.5 MW solar power plant, and they are available from the corresponding author upon request).

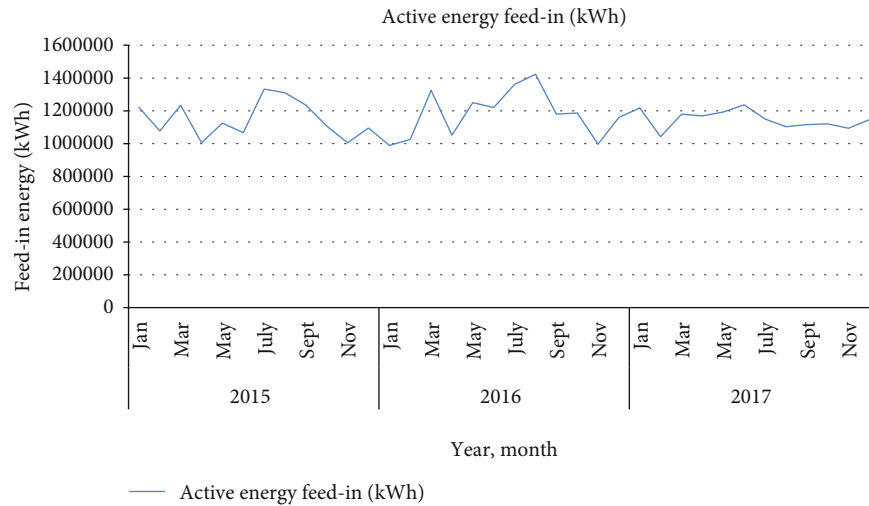


FIGURE 2: Energy feed-in in the national power grid by the Rwamagana gigawatt solar power plant (2015-2017).

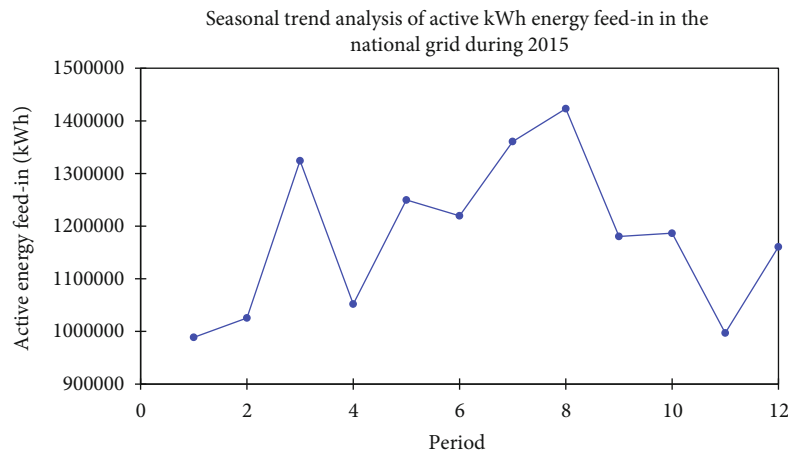


FIGURE 3: Seasonal Mann-Kendall Trend Test analysis of active kWh energy feed-in in the national grid during 2015.

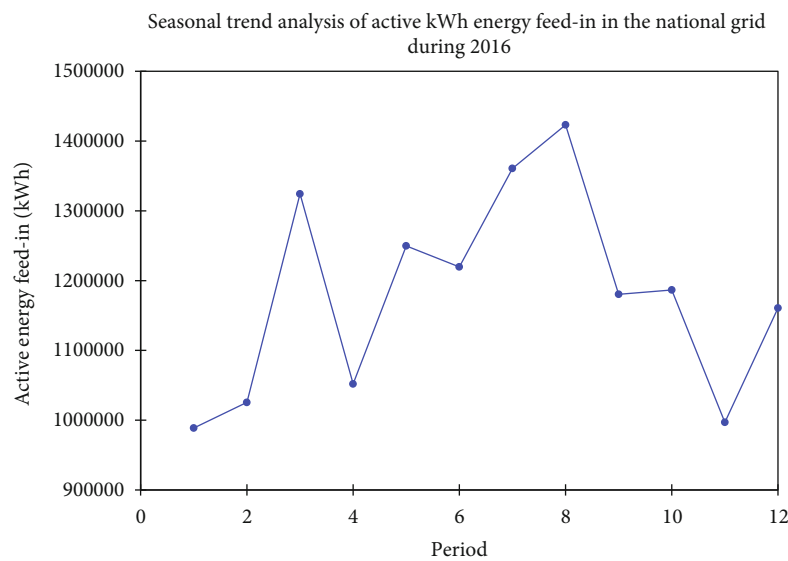


FIGURE 4: Seasonal Mann-Kendall Trend Test analysis of active kWh energy feed-in in the national grid during 2016.

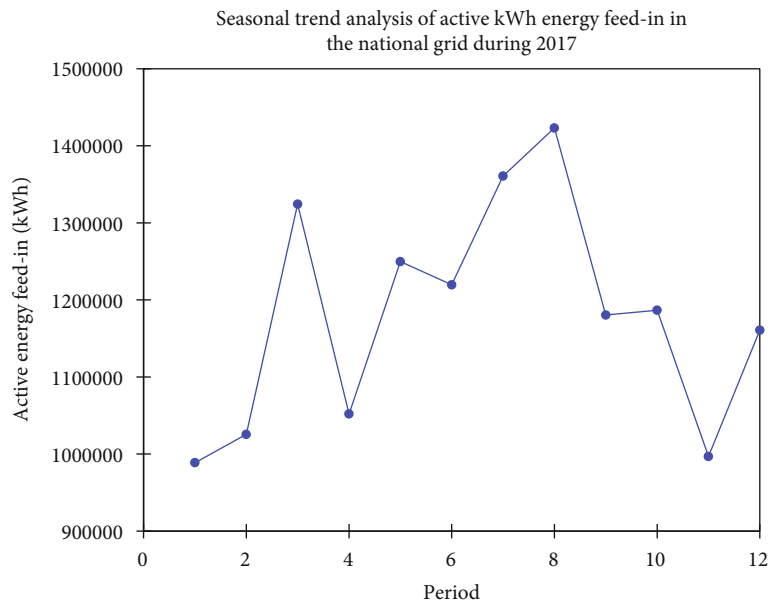


FIGURE 5: Seasonal Mann-Kendall Trend Test analysis of active kWh energy feed-in in the national grid during 2017.

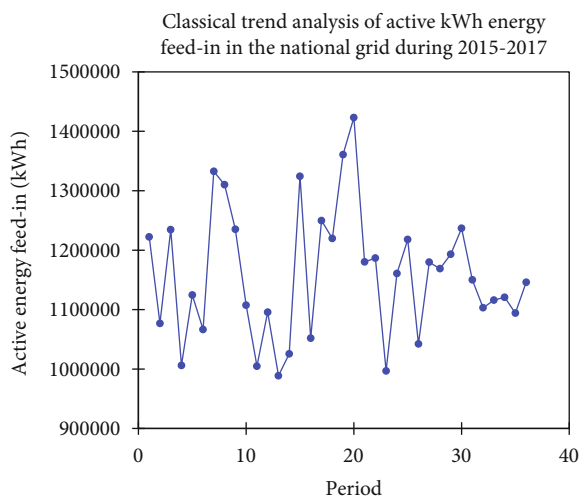


FIGURE 6: Classical Mann-Kendall Trend Test analysis of active kWh energy feed-in in the national grid during 2015-2017.

4. Modeling and Optimization of Off-Grid PV Systems

4.1. Modeling Optimization of Stand-Alone PV for an Individual Household in Rwanda. The term “off-grid users” refer to the number of users who cannot be connected to and served by the public or private utility grid [31]. Accordingly, such users access electricity through either stand-alone PV, minigrids, or microgrids [32]. This section used HOMER to model, simulate, and optimize available renewable energy generation technologies (solar and hydro), as solutions to sustainable energy access for all off-grid users in Rwanda. The simulations were carried out in Western Province of Rwanda (Rutsiro, Rwanda (1°56.3'S, 29°19.5'E)).

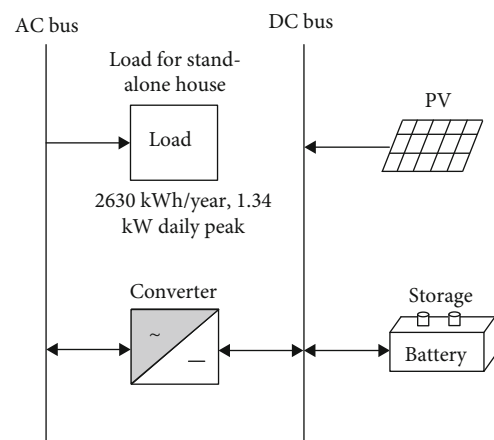


FIGURE 7: Schematic connection diagram of the stand-alone PV system.

To optimally model a stand-alone PV system, different site visits were made to Rutsiro District, Western Province of Rwanda, to a representative sample residential house. The electrical equipment, power rating, and hours of use parameters were used for analyses [33, 34].

Figure 7 is the schematic connection diagram for the stand-alone off-grid solar system. Its connection and schematic diagram include PV modules, storage batteries, converter, DC and AC buses, and electric load for an individual stand-alone solar home system. The total load profile was 2630 kWh/year (as shown in Table 1) and the daily peak load was 1.34 kW. The details of the community load of 100 households are shown in Table 2. The residential house owner listed the electrical equipment in his house and their power ratings for the purpose of determining the daily and annual electricity consumption. Further, the daily

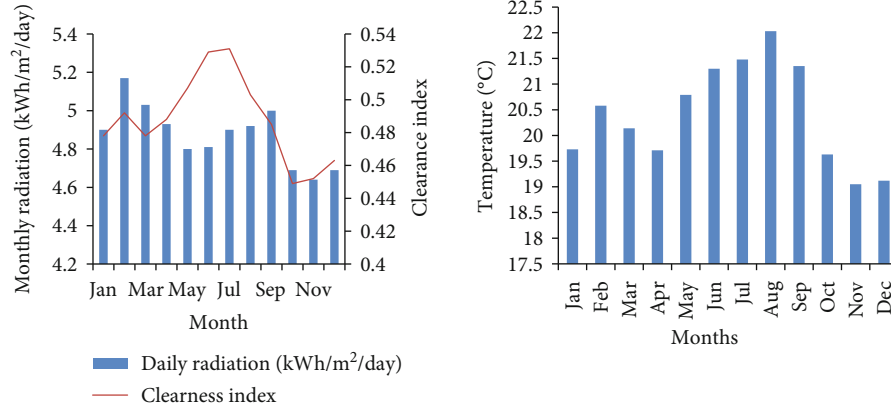


FIGURE 8: Climatic data for the case study location: (a) radiation and clearance index; (b) monthly temperature.

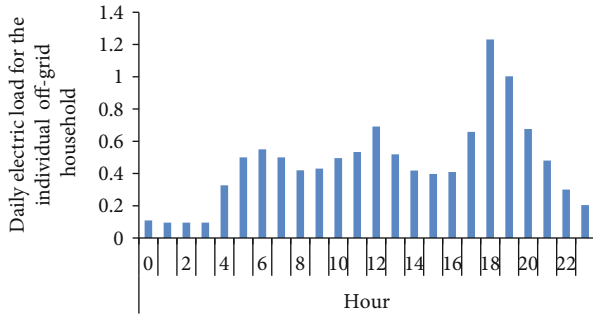


FIGURE 9: The daily load profile for the stand-alone PV system.

load was 165.44 kWh/day and the annual load was 60385.60 kWh/year for the 100 village households. The average daily load demand was 1.65 kWh/day and the average annual load demand was 603.86 kWh/year for each of the 100 village households considered. Additionally, the 7.2 kWh/day and 2630.0 kWh/year of the islanded representative PV system should be the uppermost limit of electricity consumption in the village. Thus, if the average electricity consumption between the islanded PV system and that of the 100 village households is used for design (4.43 kWh/day), much excess electricity could be produced to power micro, small, and medium enterprises in the community. This would lead to economic empowerment and healthier and cleaner communities.

Figure 8(a) shows the global horizontal irradiation and clearness index for the Rutsiro study site, while Figure 8(b) shows the monthly daily average temperature ranges in the year. The daily average radiation hovers between 4.63 kWh/m²/day and 5.17 kWh/m²/day. The clearness index is between 0.45 and 0.53, while temperature ranges between 19°C and 22°C.

Figure 9 illustrates the daily load profile for this stand-alone solar system, and the hourly load profile varied between about 0.1 kW and 1.2 kW. The daily average load was 11.26 kWh/day while the baseline was 0.47 kW with a baseline scale factor at 0.22. The peak baseline was 2.09 kW while the scaled peak power was 1.34 kW with a scale factor

of 0.22. The electric load was simulated and described with the baseline average load of 11.26 kWh/day; the baseline and scaled average of 0.47 kW and 0.3 kW, respectively; the baseline and scaled peak of 2.09 and 1.34 kW, respectively; and the baseline and scaled load factors of 0.22 and 0.22, respectively. The maximum load was about 2.5 kW for each hour throughout the year.

In PV systems having energy storage, batteries are connected for storing energy to be used by the loads once there is no irradiance or no sunshine available [35]. The storage could be customized, and cells are combined in parallel using strings to maintain storage bank voltage capacity. Their general performance varies from one model to another, but equations for their common characteristic parameters are expressed in the following equations [23, 36]:

$$V = V_0 - RI - K \left(\frac{q_{\max}}{q_{\max} - \int I dt} \right) + a e^{-B \int I dt}, \quad (15)$$

where V is the final storage voltage, V_0 is the storage voltage constant, K is the polarization voltage constant, $q_{\max}$ is the battery capacity, q is the real battery charge, a is the exponential zone of the angular distance from the east point of the horizon where the sun rises or from the west point where it sets, and B is the inverse exponential zone time constant. R is the internal resistance, I is the current of the battery, dt is time steps, and $\int I dt$ is the time-sensitive charge capacity removal function. Further, $\int I dt$ is evaluated as

$$\int I dt = q_{\max} - q. \quad (16)$$

However, the parameters V_0 , K , a , and B are supplied by manufacturers.

The quantity of charge removed depends on the Coulomb rate of battery capacity discharge. Hence, charge removal becomes [23, 36]

$$\begin{aligned} q_{\exp} &= q_{\text{full}} - I t_{\exp}, \\ q_{\text{nom}} &= q_{\text{full}} - I t_{\text{nom}}, \end{aligned} \quad (17)$$

where q_{full} is the completely charged capacity, q_{exp} (%) is the percentage of q_{full} transferred at the end of the exponential zone (%), and q_{nom} (%) is the percentage of q_{full} transferred at the end of the very small and unimportant zone (%). Also, V_{full} is the fully charged battery voltage (V), V_{exp} is the end of exponential zone voltage (V), and V_{nom} is the end of nominal zone voltage (V).

$$\begin{aligned} q_{nom,\%} &= 100 \times \frac{q_{nom}}{q_{full}}, \\ q_{exp,\%} &= 100 \times \frac{q_{exp}}{q_{full}}. \end{aligned} \quad (18)$$

The exponential zone angular distance from the east point of the horizon at which the sun rises or from the west point at which it sets (a) and other parameters are evaluated from [23, 36]

$$\begin{aligned} a &= V_{full} - V_{exp}, \\ B &= \frac{3}{q_{exp}}, \\ K &= \frac{(V_{full} - V_{nom}) + a(e^{-Bq_{nom}} - 1)(q_{max} - q_{nom})}{q_{nom}}, \\ V_0 &= V_{full} + K + RI - a. \end{aligned} \quad (19)$$

The PV system power output determined by the HOMER software [23, 36] derives from

$$P_{PV} = Y_{PV} \times f_{PV} \left(\frac{G_t}{G_{t,STC}} \right) [1 + \alpha_p(T_c - T_{c,STC})], \quad (20)$$

where Y_{PV} is the rated PV array capacity, f_{PV} is the PV loss factor (%), G_t is the PV array incident solar radiation (kW/m^2), and $G_{t,STC}$ is the incident radiation standard (1 kW/m^2). Further, α_p is the power temperature coefficient ($\%/^\circ\text{C}$), and T_c is the PV cell temperature ($^\circ\text{C}$). Also, the relationship between temperature and PV power output efficiency is given by

$$\eta_{mp} = \eta_{mp,STC} [1 + \alpha_p(T_c - T_{c,STC})], \quad (21)$$

where η_{mp} is the maximum PV array efficiency and $\eta_{mp,STC}$ is the maximum PV array efficiency at standard test conditions. Since the PV array power temperature coefficient is negative, the efficiency reduces with rising cell temperature.

The optimized simulation of the stand-alone PV solar system was conducted for 4380 hours in a year, and the life cycle cost was US\$ 20915.96. The levelized energy cost (LCOE) was US\$ 0.615/kWh. The levelized cost assesses the cost competitiveness of the stand-alone solar PV generating system that comprises all costs over the lifetime of the project. This includes initial investment, operations and maintenance, fuel cost (US\$ 0.00), and capital costs [23, 36]. Thus, LCOE enables comparison between electricity-generating technologies, costs, risks, and returns [37].

Capital spending (CAPEX) is future use prime asset acquisition. The costs are only recovered over time through depreciation. Operating expenses (OPEX) are daily general business expenses and overheads [37]. Hence, operating expenses of the stand-alone PV solar system was US\$ 683.99.

Further, the rated capacity was 6.03 kW, the mean output power was 1.09 kW, and the mean output energy delivered per day was 26.2 kWh/day. The capacity factor was 18%, the total energy production was 9547 kWh/year, and the maximum power output was 6.36 kW. In addition, the PV penetration rate was 363% and the total hours of operation were 4380 hours/year. The battery specification had 30-hour autonomy with 15 kWh nominal capacity. The usable nominal capacity was 9.01 kWh and has about 60% efficiency. The lifetime throughput was 15162 kWh, and the expected life is ten (10) years. The energy input was estimated at 1691 kWh/year while the energy output was 1356 kWh/year, indicating that the battery system was 80% efficient. The storage depletion rate was 3.79 kWh/year while the annual throughput was 1516 kWh/year. The capacity-based metrics were 100% each while the ratio between total renewable production and the load was 363%. The ratio between total renewable production and generation was also 100%. Further, one minus the ratio between total nonrenewable production and renewable peak load was 100%. The renewable peak value for the ratio between renewable generation output and the load using standard HOMER was 4129.

Further, CO_2 , CO, unburned hydrocarbons, particulate matter, SO_2 , and nitrogen oxide (NO_x) gas emissions were each zero kilogramme annually. The NPC for the stand-alone PV solar system had US\$ 12074 capital costs while the operating costs were US\$ 2023. The replacement cost was US\$ 7895 while the salvage value, which was a profit or gain, was US\$ 1076. Therefore, the total NPC was US\$ 20916. Similarly, the annualized costs for the stand-alone PV solar system for capital costs were US\$ 933.95. The annualized operating cost was US\$ 156.48 while the annualized replacement cost was US\$ 610.74. The annualized salvage value (US\$ -83.23) was a profit while there were no resource costs. The total annualized cost for the stand-alone PV solar system was US\$ 1618.00.

Figure 10 indicates both annual PV solar output power and storage battery input power of the stand-alone system. The input and output data reflect the measurements for specific dates in the year on the time axis of Figure 10. The Znshine PV-Tech ZX250(48) was the PV technology deployed for this stand-alone system. The storage battery power input ranges between -1.5 and 3.0 kW while the PV output power ranged between 0.0 and 6.4 kW. Additionally, there were Fourier series representations for both input battery power (kW) and battery charge state (%).

Figure 11 indicates that storage battery charge changes from 50% to 100%. Hundred percent (100%) battery charge occurred on most days in January while the minimum occurred on 6 January. Similarly, the battery input power ranged between -1.5 and 3.0 kW. The maximum battery input power occurred with a spike on 6 January and coincides with the minimum state of battery storage charge of

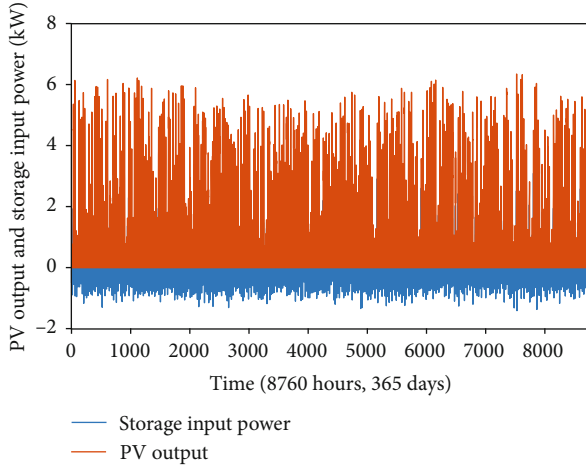


FIGURE 10: Annual PV output and input storage power for the stand-alone PV system.

50%. In addition, the minimum battery input power of -1.5 kW occurred on 23 January. The 3.0 kW maximum power battery charge was recorded on 1 January, 26 March, 30 July, and 19 November, respectively. The average battery discharge of about 1.0 kW was achieved generally throughout the year. The maximum battery discharge power of 1.5 kW occurred around 5 November.

Figure 12 reflects the annual global solar irradiation. The annual solar global radiation ranged between 0.0 kW/m² and 1.2 kW/m². The minimum and maximum annual global solar radiation occurred on 8 October and 5 November, respectively. Regarding the output power from the PV, the 0.07 kW minimum power output occurred around 12 February and 3 December, respectively. Also, the 6.5 kW maximum output power occurred in the months of February, August, September, and November, respectively. Referring to January, there were spikes and bottomed plateaus. The nearly 6.3 kW highest power output occurred on 3 and 26 January, respectively. The output of nearly 6.0 kW occurred on 9, 18, and 19 January, respectively. The minimum PV solar power output of nearly 1.0 kW occurred on 1 January. The least global solar radiation of nearly 0.07 kW/m² occurred around 28 January.

Referring to the results of PV output, solar altitude, azimuth, and angle of incidence, the PV solar power output ranges between 0.7 and 6.4 kW and the angle of incidence ranged between 0° and 92°. The 0° angle of incidence occurred on 26 March and 24 September, respectively. The 92° maximum sun incidence angle occurred in almost every sampled date throughout the year. The solar azimuth varied from -162.5° to 171° between 26 March and 24 September. The solar azimuth varied between 21° and -67° during 1 January to 26 March and 24 September to 31 December. The solar azimuth ranged between -67° and 71° in January. The angle of incidence also ranged between 0° and 92°.

The following equations highlight the dependence of PV output power on different parameters such as temperature, irradiation, azimuth, latitude, longitude, altitude, angle of

incidence, and incident radiation [23, 36, 38–41]. The solar air temperature is expressed as

$$t_e = t_o + \frac{\alpha I_t}{h_o} - \frac{\varepsilon \Delta R}{h_o}, \quad (22)$$

where t_o is the open air temperature, h_o is the surface heat transfer factor (W/m²°C), I_t is the total solar surface irradiation, α is the solar energy surface absorption factor, and $\varepsilon \Delta R/h_o$ is the correction factor. Further, solar radiation comprises straight, scattered, and combined solar radiation. Combined solar radiation is the aggregate of straight, dispersed, and the earth's radiation. The proportion of sunshine noticeable on the earth's surface depends on the eccentricities of the earth's orbit that are continuously changing distances between the sun and the earth. The quantity of sunshine beam at right angles to the earth's surface can be evaluated as

$$G_{0n} = G_{SC} \left(1 + 0.033 \times \cos \frac{360n}{365} \right), \quad (23)$$

where G_{0n} is the extraterrestrial incident radiation (kW/m²), G_{SC} is the solar multiplication factor (1.367 kW/m²), and n is the days in a year (numbered from 1 to 365).

A slope is a surface with one end higher than the other. A zero slope is horizontal while 90° slope is vertical. Azimuth is the arc of the horizon between the meridian of any place and a vertical circle passing through the sun. Thus, for simulation, zero azimuth is southwards while positive angles are westwards. Hence, -45° azimuth faces southeast and 90° azimuth faces west. Solar declination becomes

$$\delta = 23.45 \times \sin \left(360^\circ \frac{284 + n}{365} \right), \quad (24)$$

where n is the day in a year (numbered from 1 to 365).

Time affects the position of the sun and is explained by an hour angle. Hence, the hour angle is expressed as

$$\omega = (t_s - 12) \times 15^\circ, \quad (25)$$

where t_s is 12 hours at noon.

The incidence angle is the angle between the sunshine rays and the perpendicular surface. Thus,

$$\cos \theta = A - B + C + D + E, \quad (26)$$

where $A = \sin \delta \sin \phi \cos \beta$, $B = \sin \delta \cos \phi \sin \beta \cos \gamma$, $C = \cos \delta \cos \phi \cos \beta \cos \omega$, $D = \cos \delta \sin \phi \sin \beta \cos \gamma \cos \omega$, and $E = \cos \delta \sin \beta \sin \gamma \sin \omega$. Also, the incident angle is θ (°), β is the surface slope (°), γ is the surface azimuth (°), ϕ is the latitude (°), δ is the solar decline (°), and ω is the hour angle.

The zenith angle is the angle between the highest point and the centre of the sun's disc. The sun reaches its zenith at midday. Therefore, the zenith angle is zero at noon and

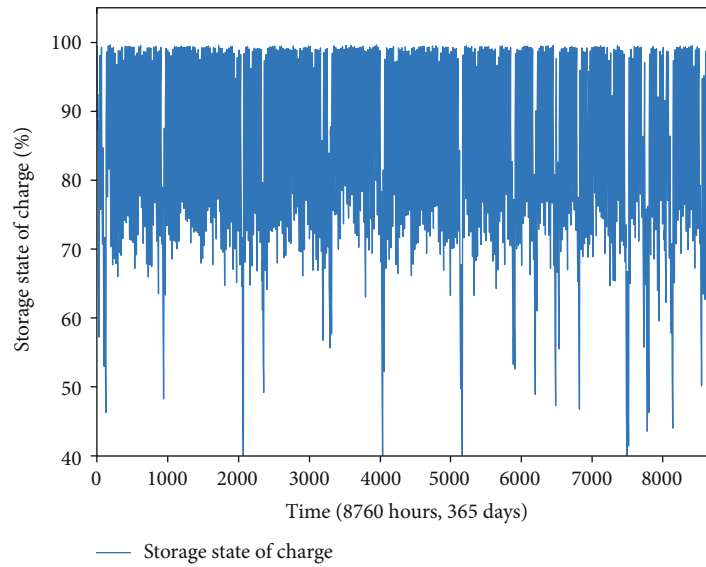


FIGURE 11: Storage charge state for the stand-alone PV system.

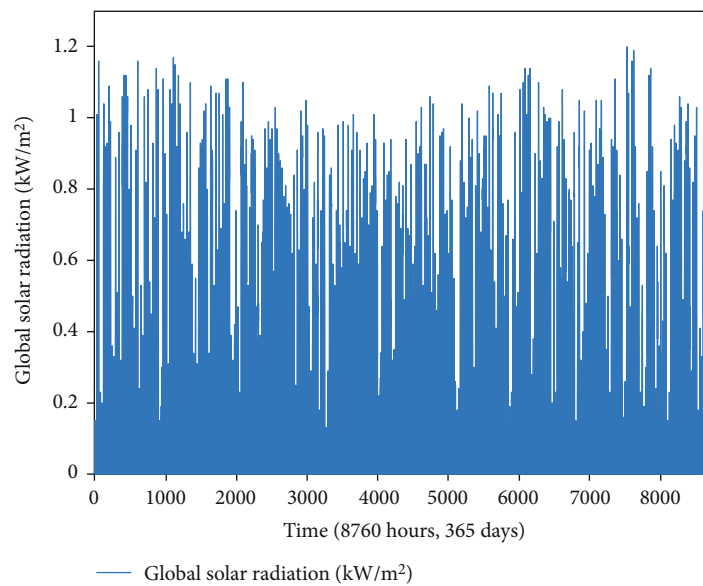


FIGURE 12: Global solar radiation.

90° at sunrise. If the surface slope angle (β) = 0° is substituted into Equation (26) above, then

$$\cos \theta_z = \cos \theta \cos \delta \cos \omega + \sin \theta \sin \delta, \quad (27)$$

where θ_z is the zenith angle ($^\circ$).

The clarity index used in HOMER PV calculation is a dimensionless quantity ranging from zero to one. It is the portion of solar radiation that passes through the atmosphere to the earth's surface. It is expressed as

$$K_T = \frac{H_{ave}}{H_{o,ave}}, \quad (28)$$

where H_{ave} is the mean monthly radiation on the earth's horizontal plane ($\text{kWh}/\text{m}^2/\text{day}$) and $H_{o,ave}$ is the extraterrestrial horizontal plane radiation at the highest earth's atmosphere ($\text{kWh}/\text{m}^2/\text{day}$).

The simulation results of the aggregate annual electric load against unmet load indicate that 0.43 kW was the minimum total electric load supplied on 2 July while the maximum of nearly 1.37 kW occurred around 5 November. There were four days of unmet loads on 26 March, 18 June, 30 July, and 5 November, respectively. The 0.35 kW highest unmet load occurred on 26 March while 0.02 kW least unmet load occurred on 18 June. It shows that at 98.9% of the time, the islanded PV solar power system supplied the needed power to customers. That means that 98.9% is the efficiency

of power supply or output sufficiency of operating the islanded PV solar system. Simulation results for January indicate that the total electric load was met. As can be seen, the required electric load was adequately supplied to the customers in January and throughout the year. Also, there were no unmet electric loads from the stand-alone PV solar power system for most days of the year.

4.2. Modeling and Optimization for Off-Grid PV Minigrad Systems

4.2.1. PV System with Storage for Off-Grid Community. Figure 13 is the illustration of the nonnetworked PV minigrad with storage. The AC community load was 60385.6 kWh/year and had a daily 20.46 kW peak. PV supplies the DC and the converter interconnects the AC with the DC supply to the storage facility.

The maximum power battery charges of the dynamic storage model [23, 36] used in modeling and simulation are

$$\begin{aligned}
 P_{\text{batt,cmax,kbm}} &= \frac{kQ_1 e^{-k\Delta t} + QkC(1 - e^{-k\Delta t})}{1 - e^{-k\Delta t} + C(k\Delta t - 1 + e^{-k\Delta t})}, \\
 P_{\text{batt,cmax,mcr}} &= \frac{(Q_{\text{max}} - Q)(1 - e^{-\alpha_c \Delta t})}{\Delta t}, \\
 P_{\text{batt,cmax,mcc}} &= \frac{N_{\text{batt}} \times I_{\text{max}} \times V_{\text{nom}}}{1000}, \\
 P_{\text{batt,cmax}} &= \frac{\min(P_{\text{batt,cmax,kbm}} \times P_{\text{batt,cmax,mcr}} \times P_{\text{batt,cmax,mcc}})}{\eta_{\text{batt,c}}},
 \end{aligned} \quad (29)$$

where $P_{\text{batt,cmax}}$ is the maximum charge battery power, Q_1 is the accessible initial energy storage (kWh), and Q is the initial sum total energy storage (kWh). Further, C is the electrical storage capacity factor, k is the battery storage gauge (h^{-1}), Δt is the time (h), $P_{\text{batt,cmax,mcr}}$ is the maximum charge battery power measure, and α_c is the maximum storage charge rate (A/AH). Also, Q_{max} is the maximum size of the bank storage (kWh), N_{batt} is the battery number, I_{max} is the maximum storage charge current (A), V_{nom} is the initial storage maximum fixed voltage (V), and $\eta_{\text{batt,c}}$ is the efficiency of charge storage.

Hence, the power discharge becomes

$$\begin{aligned}
 P_{\text{batt,dmax,kbm}} &= \frac{-kCQ_{\text{max}} + kQ_1 e^{-k\Delta t} + QkC(1 - e^{-k\Delta t})}{1 - e^{-k\Delta t} + C(k\Delta t - 1 + e^{-k\Delta t})}, \\
 P_{\text{batt,dmax}} &= \eta_{\text{batt,d}} \times P_{\text{batt,dmax,kbm}}.
 \end{aligned} \quad (30)$$

Figure 14 shows the daily load profile of the solar PV minigrad with storage. The minimum electric load of 2.0 kWh was required in the first five hours of the day (0-4 hours), and the 12.0 kWh maximum community electrical loads occurred between 18 and 21 hours (inclusive). There was a constant electric load of about 8 kWh from 8 to 15 hours (inclusive).

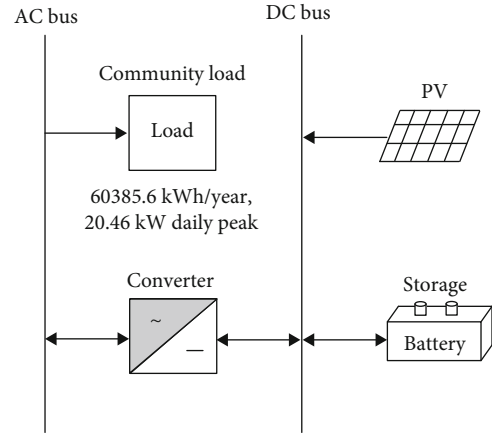


FIGURE 13: Sketch of the nonnetworked PV minigrad with storage.

Figure 15 represents the annual PV output for the off-grid PV minigrad with storage. The 10 kW minimum PV power outputs occurred on 12 February, 21 May, 8 October, 5 November, and 3 December. Also, the nearly 80.0 kW maximum PV power outputs occurred twice between 5 and 19 November. Referring to the simulation results for January, the minimum PV solar power output was nearly 10.0 kW while the maximum was nearly 80.0 kW. The global solar power output spikes were each bottomed around 0.0 kW/m² and peaked at nearly 1.17 kW/m². Majority of both the global solar power output and generic flat plate PV output power spikes were each greater than 50.0 kW and 0.8 kW/m², respectively. Regarding the storage, the battery storage state of charge was fully charged at 100% for most of the year. Although the annual large battery storage state of charge varied between 40.0% and 100.0%, the least battery storage state of charge of about 40.0% occurred around 31 March, 7 August, and 24 November, respectively. Further, the storage input power varied between -20.0 kW and 76.0 kW. The majority of the storage input power hovered between -20 kW and 40.0 kW. The highest storage input power peaked nearly 76 kW and occurred around 12 November. The other relatively high storage input power of nearly 60.0 kW occurred around 8 January, 12 February, 28 March, and 7 December.

Figure 16 reflects the graph of annual storage against storage discharge power for the off-grid PV minigrad with storage. The storage charge power ranged between 7.0 kW and 72.0 kW. Although there was a concentration of storage charge power around 25.0 kW, there were so many spikes with the highest about 72.0 kW occurring around 12 November. Hence, storage charge power above 50.0 kW occurred on 2 January, 8 January, 14 January, 12 February, 28 March, 19 May, 20 June, 4 August, 17 September, 30 September, 12 October, 2 November, 11 November, 25 November, and 27 December. Conversely, the storage discharge power hovered between 0.0 kW and 20.0 kW while the majority of the storage discharge power was around 12.0 kW. Also, the annual storage discharge power was rather relatively stable and the dispersion or risk of operation was much lower when compared with the storage charge power. This is so because the

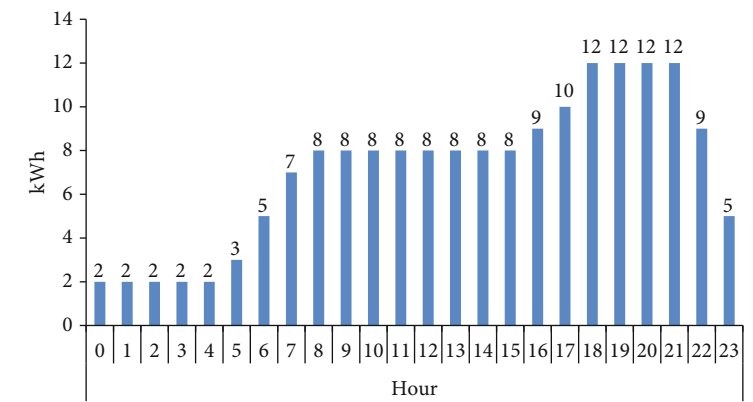


FIGURE 14: Daily load profile of the PV solar minigrid with storage.

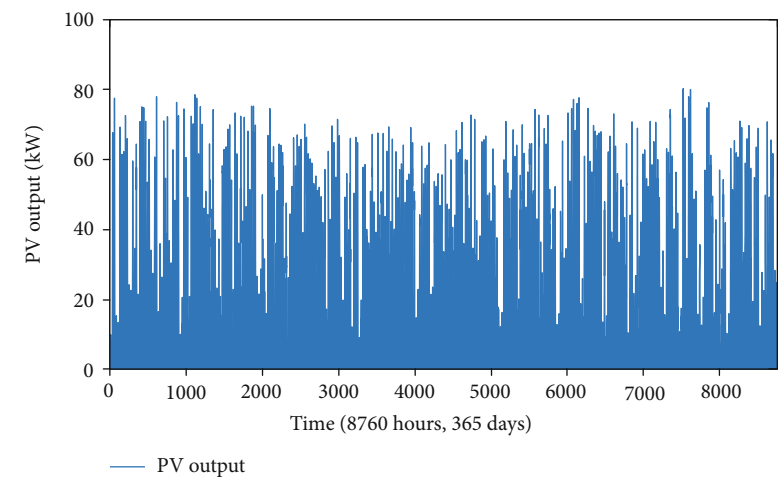


FIGURE 15: PV power output of the off-grid PV minigrid system with storage.

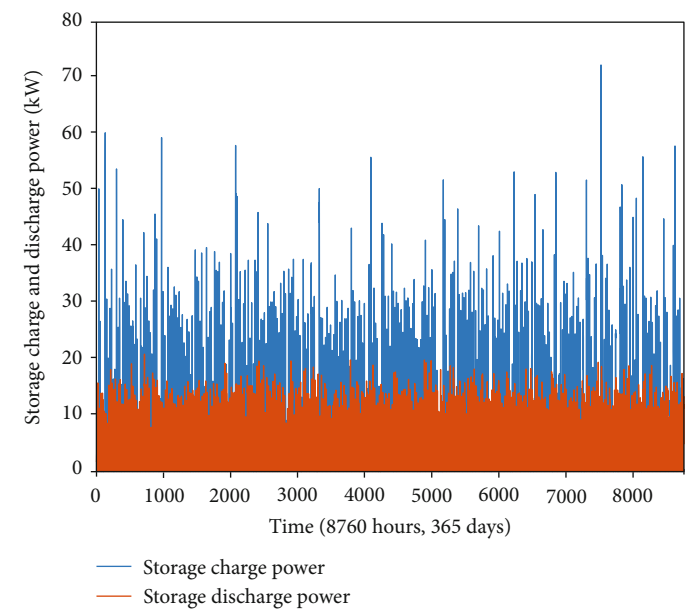


FIGURE 16: Power storage charge against power storage discharge of the unconnected PV minigrid with storage.

risk is measured by the level of variability or range between the lowest and highest operational levels or values.

Regarding the capacity to satisfy the load, the results indicate that the electricity produced by the off-grid PV minigrid with storage was excess on most days of sampled data. Also, many days show deficit in electricity for the off-grid PV minigrid system with storage. Simulation results show pattern spikes in excess electricity production and a large area at the bottom of the wave transformations without excess electricity production or even deficits. There were no deficits or no excess electricity production for 1, 4-6, 11-13, and 16-30 January. The striking pattern of the spikes occurred at the centres between the days in the histogram grid lines. This implies that the maximum electricity production and by extension the excess electricity production for the off-grid PV minigrid system with storage usually happened around midday, when the sun rays are overhead. Regarding the annual total electricity load saved against renewable penetration rate, the highest renewable penetration rate was around 2700.0% which occurred in January. The general renewable penetration rate about 200.0% was witnessed throughout with pockets of relatively larger renewable penetration rates over 1000.0% occurring in January to June and August to early November. The average total electricity served was about 15.0 kW while the largest was 20.0 kW for the year.

That means that the storage charge power at any instant can be found anywhere from 7.0 kW and 72.0 kW which is a range of 65.0 kW while the storage discharge power can vary from 0.0 kW to 20.0 kW and the risk of operation is the range of 20.0 kW. It follows that the storage discharge power can be more easily controlled than the storage charge power. For the simulation results in January, the storage maximum discharge power was relatively stable for the month of January at about 170.0 kW except for the three troughs or dips or bifurcations on 5, 6, and 13 January. The lowest dip or point of bifurcations occurred on 6 January while the least of about 150.0 kW occurred on 5 January. In contrast, the Fourier series representations of the storage discharge power are spiked half-wave rectified signals with distortions at the trailing edge. These can be seen as DC signals with distortion and attenuation.

In order to get better analysis and understanding on the storage of this nonnetworked minigrid, the simulation results were deeply viewed and there were Fourier series of signal input power against charge storage. The charge storage condition hovered between 10.0% and 100.0% while the storage input power varied between -10.0 kW and 60.0 kW. The major dips or bifurcations for the storage state of charge occurred on 2 January (30.0 kW), 6 January (10.0 kW), and 13 January (20.0 kW), and dips around 37.0 kW occurred on 12 January, 17 January, 25 January, 28 January, and 30 January, respectively. Similarly, the peaks above 40.0 kW for the storage input power occurred, respectively, on 2, 6, 13, 17, and 25 January. Simulation results depict the maximum storage charge power superposed on the maximum storage discharge power for January. The maximum discharge power was stable around 170.0 kW for the month except for three days with dips or bifurcations on 5, 6, and 13 January, respectively. Conversely, the storage maximum charge power

varied from 0.0 kW to about 130.0 kW. These were a combination of distortions and attenuations with varying proportions of signals on each day that oscillates without a readily visible pattern. Also, spikes greater than 50.0 kW occurred on 2, 6, 9-13, 17, 23-25, 28, and 30 January. The spikes were relatively stable except where the bifurcations on 6 and 13 January were large and have corresponding spikes of maximum storage charge power.

Regarding the renewable energy penetration against total electricity load served, the largest total electrical load served ranged between 1.0 kW and 20.0 kW, with the majority around 12.5 kW. The highest penetration rate occurred on 7 January while the least about 150.0% occurred on 1 January. The simulations indicate that there were no unmet electrical loads for January while the served total electrical load varied between 1.0 kW and 18.0 kW. The highest electrical load served occurred on 23 January while the least occurred on 2 January. Conclusively, the total net present cost (NPC) was US\$ 304482.30, CAPEX was US\$ 204878.00, and OPEX was US\$ 7705.00. Further, the levelized COE/kWh was US\$ 0.390, the fuel consumption was US\$ 0.00, and the internal rate of return (IRR) was 28.1% while the discounted payback year and simple payback year were 5.98 years and 4.92 years, respectively. Additionally, the rated capacity was 84.6 kW, the mean output power was 13.8 kW, and the mean output energy per day was 330 kWh/day. The capacity factor was 16.2%, the total production was 120451 kWh/year, and the maximum output power was 80.3 kW while the PV penetration rate and hours of operation were, respectively, 199.0% and 4380 hours. Further, the battery autonomy was 53 hours, the nominal capacity was 609 kWh, and the usable nominal capacity was 366 kWh (60.0% efficiency). The lifetime throughput was 344000.00 kWh, the expected lifespan was 10 years, the energy in was 38297 kWh/year, and the energy out was 30769 kWh/year (80.3% efficiency).

This implies that the energy conversion rate or performance efficiency was 80.3%. The storage depletion was 147 kWh/year while the annual throughput was 34401.00 kWh/year. However, both the ratio between nominal renewable capacity and total nominal capacity and the ratio between usable renewable capacity and total capacity were 100.0%, respectively. The ratio between total renewable production and load, ratio between total renewable production and generation, and one minus ratio between total nonrenewable production and load were 200%, 100%, and 100%, respectively. The HOMER standard peak values of the ratio between renewable output and load, ratio between renewable output and total generation, and one minus ratio between nonrenewable output and total load were 2663%, 100%, and 100%, respectively. Similarly, the gas emissions for CO₂, CO, unburned hydrocarbons, particulate matter, SO₂, and nitrogen oxides (NO_x) were each zero kilogramme a year, respectively. The NPC comprise capital costs of US\$ 204878.00, operating costs of US\$ 61008.00, replacement costs of US\$ 46228.00, salvage value of US\$ -7632.00 (or gain US\$ 7632.00), zero resource costs, and total net present costs amounting to US\$ 304482.00. Nevertheless, the annualized costs comprise US\$ 15848.00 capital costs, US\$ 4719.00 operating costs, US\$ 3576.00 replacement costs, US\$ -590.35 salvage value

(gain or profit at the end of service life), and zero resource costs (US\$ 0.0/kWh/year) while the total annualized costs become US\$ 23553.00.

4.2.2. Hybrid PV-Hydro with Storage Off-Grid Community Load. Figure 17 reflects the schematic diagram of the off-grid hybrid solar PV-hydro minigrid with storage facility. The hydro turbine system provides AC load to the community (60385.6 kWh/year and daily peak of 20.46 kW) while the PV system provides DC power to the community. The converter changes either AC/DC or DC/AC from either the hydro turbine or PV system as the occasion demands.

Figure 18 indicates the river stream level for the off-grid hybrid PV solar hydro minigrid with storage. The flow rate ranged between 200 and 390 litres/second. The lowest flow rate of 390.0 l/s occurred in April and November, respectively. These periods also corroborate the two rainy seasons observed annually in Rwanda [13].

The quantity of water passing through the hydro turbine is its flow rate. It is expressed as [23, 36]

$$Q_{\text{turbine}} = \begin{cases} 0, & \text{if } Q_{\text{available}} < Q_{\text{min}}, \\ Q_{\text{available}}, & \text{if } Q_{\text{min}} \leq Q_{\text{available}} \leq Q_{\text{max}}, \\ Q_{\text{max}}, & \text{if } Q_{\text{available}} > Q_{\text{max}}, \end{cases} \quad (31)$$

where $Q_{\text{available}}$ is the accessible water flow to the water wheel (m^3/s), Q_{min} is the minimum flow rate of the water wheel (m^3/s), and Q_{max} is the maximum flow rate of the water wheel (m^3/s).

Hydropower generation rises with distance of a vertical lift or vertical drop and quantity of water flow rate. The highest and optimal hydropower is produced when frictional loss in pipes is a third of static head. Further, power generation and water use depend on nozzle sizes with suitable filtration.

Figure 19 shows the graph of PV output power and hydro turbine power. The hydropower output was constant around 440.0 kW until towards 30 November when the output became 0.0 kW until the end of the year. Conversely, the stream flow ranged between 200 l/s and 400 l/s with each monthly stream flow rate determined at constant average values. The highest flow rates of 400 l/s agreed with the two rainy seasons in Rwanda that usually peak in April and November, respectively. Regarding the radiation, the global solar insolation ranged between $0.17 \text{ kW}/\text{m}^2$ and $1.2 \text{ kW}/\text{m}^2$ and the global solar insolation averaged around $0.9 \text{ kW}/\text{m}^2$ while the PV solar power output peaked around 170.0 kW in January to March and July to early November. The peaks also occurred around midday while the PV output power was equally at the tips of each spike. The PV power output bottomed out around midnight when there was no sunlight and the transition between the previous day and the next. The peaks occurred around 3, 9, 18, 19, and 26 January, respectively; the solar global insolation ranges between $0.17 \text{ kW}/\text{m}^2$ and $1.2 \text{ kW}/\text{m}^2$. The global solar insolation of over $1.0 \text{ kW}/\text{m}^2$ occurred between January and 12 March and between 13 August and 31 December. The incident angle and universal flat plate PV solar altitude ranged between 0° and 100° , respectively. The universal flat plate PV solar

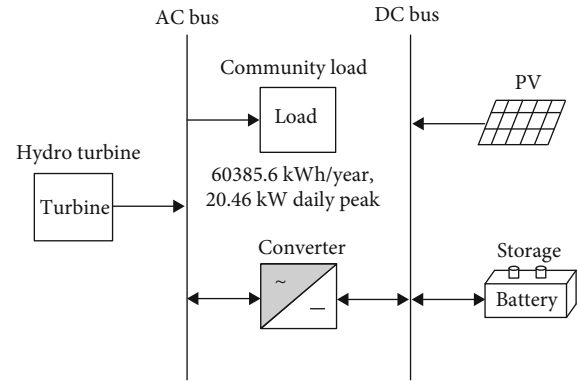


FIGURE 17: Schematic diagram for the off-grid hybrid PV and hydro minigrid system with storage.

azimuth varied between -170° and 100° from 1 January to 12 March and from 24 September to 31 December. Further, it ranged between -130° and 170° from 26 March to 24 September. The generic flat plate PV incident solar angle ranged between -30° around 30 July and about 130° around 10 September.

Figure 20 indicates the graph of annual charge storage for the hybrid solar PV-hydro nonnetworked minigrid system. The majority storage state of charge varied between 90% and 100% from 1 January up to the end of November. The storage state of charge reduced to as low as 70% around 3 December and oscillated between 80% and 100% from then onwards until 31 December. Although the input power hovered around -5 kW and $+5 \text{ kW}$ from 1 January until 3 December, the storage input power changed wildly between -20 kW and 45 kW from 3 to 31 December. This also shows that the demand for stored input power was high as well as the necessary electricity production capacity to match the shortfall.

Regarding the annual storage charge power and storage discharge power for the hybrid PV solar hydro off-grid minigrid system, the storage discharge power ranged between 0.0 kW and 7.0 kW from 1 January up to 3 December. The storage discharge power peaked around 20.0 kW after 3 December and was going beyond 10.0 kW for the remaining part of the year. Similarly, the storage charge power hovered around 0.0 kW and 5.0 kW for the period of 1 January up to 3 December. The storage charge power peaked about 45.0 kW between 3 and 31 December with majority of the storage charge exceeding 12.0 kW . There was a concentration and much activity for the off-grid hybrid PV solar hydro minigrid system through much storage power usage and the correspondingly high storage discharge power experienced. These increased activities during the period of 3 to 31 December could be attributed to end-of-year festivities in the community.

For annual maximum power storage charge and maximum power storage discharge of the islanded hybrid solar PV-hydro minigrid, the maximum power storage charge varied between 0.0 kW and 12.0 kW from 1 January to 3 December. The much higher values of the maximum storage charge power over 12.0 kW and peaking at 41.0 kW occurred around

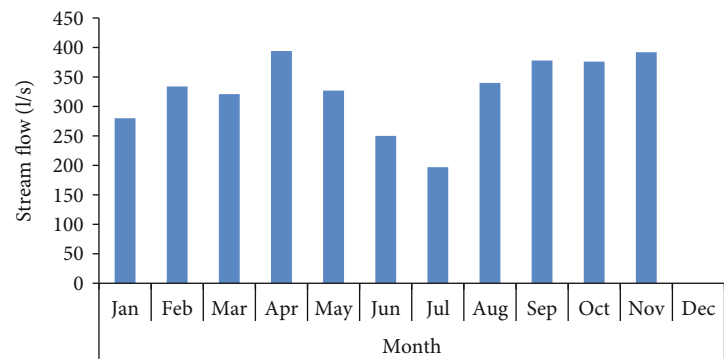


FIGURE 18: River flow levels of the nonnetworked hybrid PV-hydro minigrid system with storage.

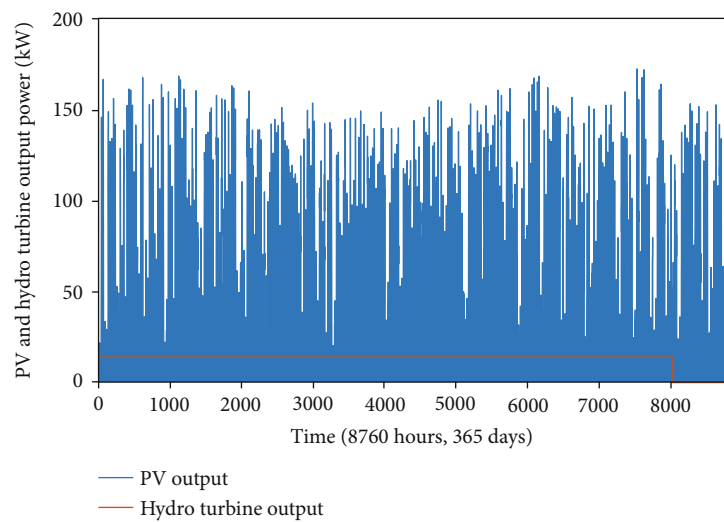


FIGURE 19: Hybrid PV-hydro turbine power output for the minigrid system.

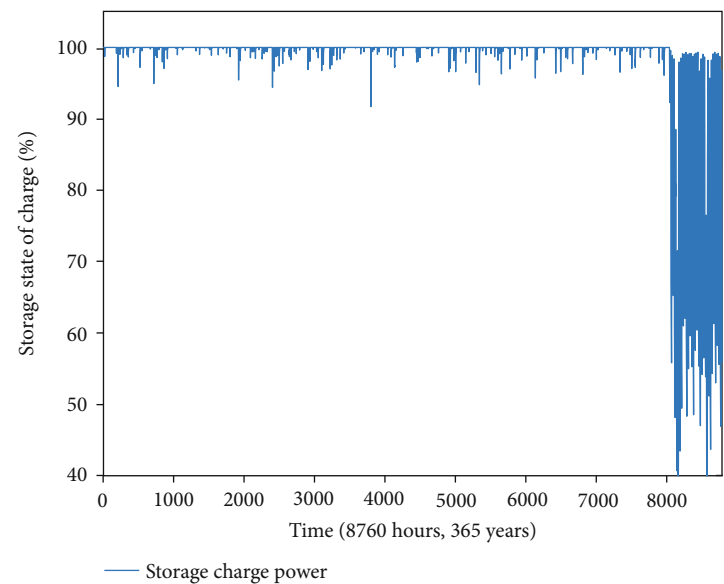


FIGURE 20: PV storage charge state of the hybrid PV-hydro islanded minigrid system.

5 December. Conversely, the maximum storage discharge power peaked about 53.0 kW throughout the year. The maximum storage discharge power reduced to 0.0 kW around 5 and 24 December. Also, the maximum storage discharge power discharged massively around 50.0 kW to 0.0 kW from 3 to 31 December during that one-month period. Regarding the total electricity load served against renewable penetration rate for December, the total electrical load indicates oscillatory waveforms of increasing and decreasing spiked distortions and attenuations with irregular repeatability. The total electrical load varied between 0.0 kW and 18.0 kW. The majority of the total electrical load laid around 10.0 kW. Additionally, the renewable penetration rate ranged between 400% and 2800%. The renewable penetration waveforms bottomed around 0% on many days in December. The analysis of annual graphical waveforms of excess electricity production against unmet electrical load shows that there were very few unmet electrical loads for the year. It also shows that slight unmet loads were recorded about 2 or 3 days in December. Additionally, the excess electricity production peaked around 180 kW while the minimum was 0 kW only in the days of the unmet electrical loads in December.

In summary, the total net present cost (NPC) was US\$ 564779.10, CAPEX was US\$ 507589.00, OPEX was US\$ 4424.00, levelized COE/kWh was US\$ 0.7239, and fuel consumption was zero litre per year. Further, the PV system has 182 kW rated capacity, the mean output power was 30.7 kW, the mean output energy was 736 kWh/day, and the capacity factor was 16.9%. The total annual energy production was 268780 kWh/year; maximum output power, 172 kW; and PV penetration rate, 445%. It operates 4380 hours/year with US\$ 0.0904/kWh levelized costs. The battery autonomy is 17.5 hours with 201 kWh nominal capacity and 121 kWh usable nominal capacity (60.2% efficiency). The lifetime throughput is 32211 kWh, the expected life is 10 years, and the energy input was 3502 kWh/year while the energy output is 2881 kWh/year (82.3% efficiency). The storage depletion rate is 88.7 kWh/year while the annual throughput is 3221 kWh.

Nevertheless, the hydro system contributes 11.0 kW nominal capacity, 12.8 kW mean output power, 117% capacity factor, and 112294 kWh/year total annual energy production. The maximum output power is 14.0 kW; hydro penetration rate, 186%; operational hours, 8016 hours/year; and levelized cost, US\$ 0.0158/kWh. The ratio between nominal renewable capacity and total nominal capacity and ratio between usable renewable capacity and total capacity was 100%, respectively. Also, the ratio between total renewable production and load, ratio between total renewable production and generation, and one minus ratio between total nonrenewable production and load were 631%, 100%, and 100%, respectively.

Similarly, the peak values for the HOMER standard of the ratio between renewable output and load, ratio between renewable output and total generation, and one minus ratio between nonrenewable output and total load were 9683%, 100%, and 100%, respectively. Additionally, gas emissions for CO₂, CO, unburned hydrocarbons, particulate matter, SO₂, and nitrogen oxides (NO_x) were each zero kilogramme

in a year, respectively. The project cash flow for the net present costs (NPCs) comprises US\$ 507589.00 capital, US\$ 44398.00 operating costs, US\$ 15944.00 replacement costs, US\$ -3151.00 salvage value (gain or profit at the end of the plants' useful life), US\$ 0.00 resource costs, and US\$ 564779.00 total costs (less salvage value). The annualized costs include US\$ 39264.00 capital, US\$ 3434.00 operating costs, US\$ 1233.00 replacement costs, US\$ -243.78 salvage value (or profit), and US\$ 0.00 resource costs while the total annualized costs become US\$ 43688 (less salvage value).

4.3. Summary. Section 4 contains the modeling and optimization of islanded PV systems. Also, parameters from storage and PV system models were discussed. Through the HOMER built-in optimizer, with its derivative-free methods, three PV systems with storage (islanded PV system of individual household load, an off-grid PV minigrid with storage, and a hybrid of hydropower and PV minigrid system with storage) were, respectively, modeled and analyzed for the purpose of obtaining optimal energy systems which can help in solving key technology energy issues for Rwanda. The islanded PV system model showed that it can annually produce excess electricity of 6445 kWh (67.5%), having unmet electric load of 0.649 kWh/year (0.0247%) and capacity shortage of 2.54 kWh (0.0965%). The system was completely renewable energy technology (100% renewable fraction) with maximum renewable penetration of 4129% and US\$ 0.6155 LCOE and zero hydrocarbon and gas emissions. For the off-grid community load, two minigrid systems were modeled and analyzed. The model PV minigrid system with storage indicated 51348 kWh/year (42.6%) excess electricity generation, 42.9 kWh/year (0.0711%) unmet electric load, 56.4 kWh/year (0.0934%) capacity shortage having 100% renewable fraction and 2663% maximum renewable energy penetration rate. The system also presented zero carbon and gas emissions and US\$ 0.3903 LCOE. The hybrid of the hydropower and PV minigrid system with storage presented the excess electricity of 319791 kWh/year (83.9%), the unmet electric load of 38.6 kWh (0.064%), the capacity shortage of 60.3 kWh/year (0.999%), the renewable fraction of 100%, the maximum renewable penetration of 9683%, US\$ 0.7239 LCOE, and zero carbon and gas emissions. In this section, the results showed that PV minigrids with storage can be the optimal and affordable solution for electrification and energy access acquisition in off-grid areas of Rwanda.

5. Modeling and Optimization of Grid-Connected PV Systems

5.1. PV System as a Solution to an Unreliable Grid. Figure 21 shows the power outages in Kigali City which contains the number of respondents against duration ranges of blackouts in minutes. About 328 respondents indicate that they witnessed blackouts lasting up to 100 minutes and about 75 respondents indicate that they have witnessed blackouts lasting between 100 and 200 minutes while about 10 respondents have also witnessed blackouts lasting between 200 and 300 minutes. Around 23 respondents have experienced blackouts lasting between 301 and 400 minutes while about 8

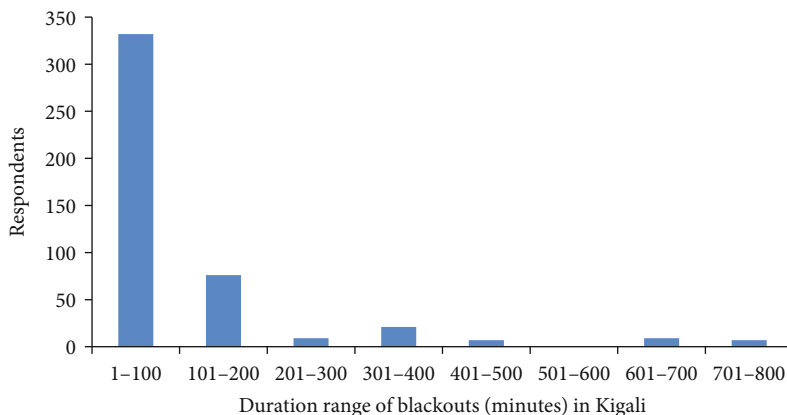


FIGURE 21: Power outage in Kigali City (authors' calculation based on living household conditions, EICV 5, 2017).

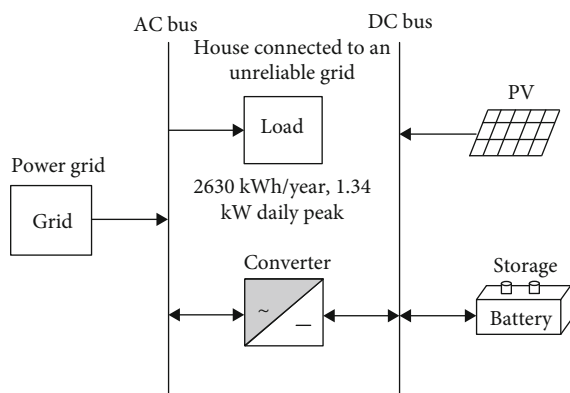


FIGURE 22: Schematic diagram for the unreliable grid with the PV system solution.

respondents have witnessed between 401 and 500 minutes of electricity blackouts. Although 10 respondents indicate experiencing electricity blackouts between 601 and 700 minutes, about 8 respondents have experienced between 701 and 800 minutes of power failure earlier in Kigali City. In the last two hundred years, electrical power is gradually replacing food as the prime necessity of life in modern societies. Modern business, commerce, and industry rely on affordable, reliable, and sustainable electricity supply to perform their daily functions. Thus, loss and absence of electricity for any length of time has presented untold hardships, decay, death, and inconveniences to peoples of all ages and creeds. This is so because electricity has become the prime driver of almost every facet of life activities.

Figure 22 is a sketch of unreliable grid-PV system solution containing an AC grid, electric load, PV system, converter, and storage. The electric load was 2630 kWh/day and had daily 1.34 peak load while the converter changes AC to DC and the other way round, to and from the storage. The design considered unreliable grid parameters such as random outages, scheduled outages, and grid hourly outage schedules. The grid reliability chart indicates mean outage frequency per year of 12.00 and mean repair time variability of 6 hours. The grid outage chart relates to blackouts for each

day of the year coupled with the corresponding periods where outages would be most likely and when to carry on preplanned outages to maintain power system balance. Regarding the change of PV power output against global solar insolation for the unreliable grid model, the PV solar power output varied between 1.0 kW and 8.5 kW while the least global solar insolation was about 0.15 kW/m² and the largest was 1.2 kW/m². The average global solar insolation was around 5.0 kW/m² while the average global insolation was 0.8 kW/m².

Figure 23 shows the graph of annual storage state of charge for the unreliable PV-grid model using the PV system solution. The lowest storage state of charge varied between 40% and 100%. The majority of storage state of charge was over 50% all year round. The lowest storage charge occurred around 2 February, 2 June, 22 July, and 24 October, respectively. The least storage charge power output was around 0.3 kW and occurred around 2 June; the highest storage state of charge power was 1.33 kW and occurred around 10 January, 31 January to 2 March, 23 March, 2 June, 5 to 10 July, five days of sampled data points between 22 July and 2 August, and 24 October, respectively. Further, majority of the storage discharge was greater than 0.4 kW on the average throughout the year. The storage discharge power varied between 0.33 kW and 1.34 kW while concentrations of relatively higher discharge power occurred from January ending to 2 March and 6 July to 2 August. The storage input power varied between -1.3 kW and 1.3 kW. The -1.3 kW which was the least storage input power occurred around 23 July while the highest storage input power occurred around 10 January, 31 January to 2 March, 2 June, 5 to 10 July, five sampled points from 22 July to 2 August, and 24 October, respectively.

The maximum storage discharge plateaued around 1.6 kW while the least maximum storage discharge power about 1.34 kW. The band of the least maximum storage discharge power occurred on 10 January, around 3 February to 2 March, and about twenty points from 9 July to 1 September and another five data points from 2 October to 29 November, respectively. The maximum storage charge power peaked around 1.34 kW and also had each of their peak values corresponding to the least maximum storage

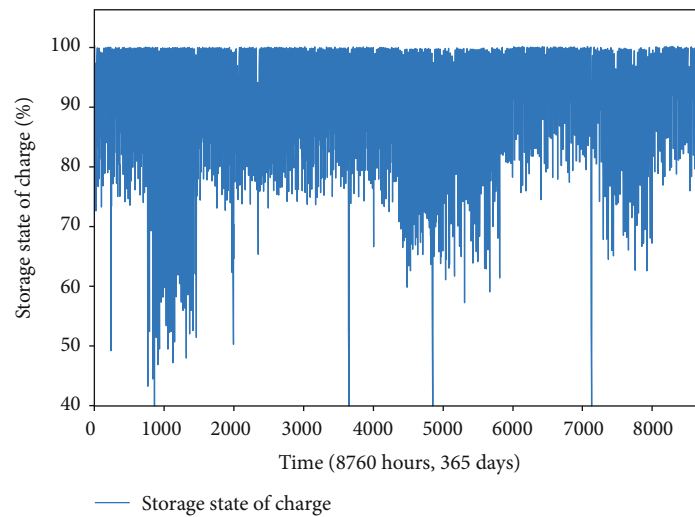


FIGURE 23: Storage input power for the unreliable grid model with the PV system solution.

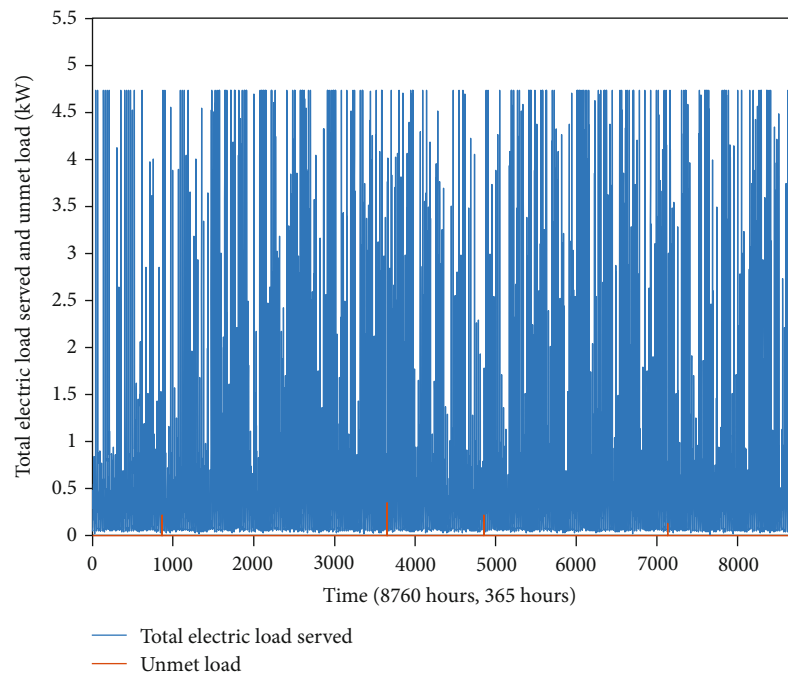


FIGURE 24: Total electric load served and unmet electric load for the unreliable grid model with the PV system as the solution.

discharge power. The lowest maximum storage charge power was 0.5 kW and occurred around 25 September and 20 October, respectively.

Figure 24 depicts the served annual aggregate electrical load plotted against unsatisfied electrical load. The highest total electrical load served was around 4.7 kW while the least was about 0.5 kW. Although the highest total electrical load of 4.7 kW was supplied most days of the year, the lowest total electrical load served (0.5 kW) occurred about 22 December. Further, the graph shows four unmet electrical loads that occurred around 6 February, 2 June, 21 July, and 25 October, respectively. The highest unsatisfied electrical load was around 0.4 kW while the least unsatisfied electrical load was about 0.13 kW. The highest total renewable output power

of about 8.43 kW occurred around 5 April while the least total renewable power output of about 0.5 kW occurred around 9 February, 21 March, 17 June, and 13 October, respectively. Conversely, the highest renewable penetration rate was 3500.0% while the least penetration rate was about 250%. The highest total renewable power output was around 8.4 kW while the least was around 1.0 kW. The least total renewable power output occurred around 10 February and 20 May, respectively, while the highest total renewable output power materialized on 10 and 14 November, respectively. The largest excess electrical production was around 7.74 kW and occurred around 7 September while the least excess electrical production of around 1.0 kW occurred about 20 May, respectively.

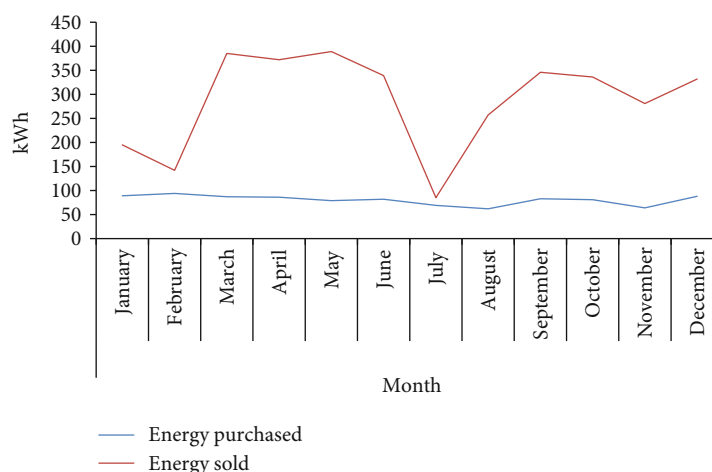


FIGURE 25: Grid sales against grid purchases of the unreliable grid with the PV system model solution.

In sum, the ratio between nominal renewable capacity and total nominal capacity and ratio between usable renewable capacity and total capacity was each 100.0%, respectively. The ratio between total renewable production and load, ratio between total renewable production and generation, and one minus ratio between total nonrenewable production and load were 206.0%, 92.9%, and 100.0%, respectively. The HOMER standard ratio between renewable output and load, ratio between renewable output and total generation, and one minus ratio between nonrenewable output and total load were 4645.0%, 100.0%, and 100.0%, respectively. Similarly, the carbon dioxide emitted was 609.0 kg/year, the sulphur dioxide emitted was 2.64 kg/year, and the nitrogen oxide emitted was 1.29 kg/year. Additionally, there was no carbon dioxide, unburned hydrocarbons, or particulate matter emitted for the PV solar system model solution. Further, the PV solar system was rated 7.94 kW, the mean output power was 1.43 kW, and the mean output energy was 34.4 kWh/day while the capacity factor was 18.1%. The total production was 12565 kWh/year; maximum output power, 8.37 kW; PV penetration rate, 478.0%; operation time, 4380 hours/year; and levelized cost, US\$ 0.0296/kWh. However, the battery autonomy was 12 hours and the nominal capacity was 6.0 kWh, while the usable nominal capacity was 3.6 kWh, indicating 60.0% efficiency. The throughput lifetime was 4800 kW, the expected life was 7.2 years, the energy input was 774.0 kWh/year, and the energy output was 596.0 kWh, indicating 80.0% efficiency. The storage depletion rate was 1.2 kWh/year, and the annual throughput was 667.0 kWh/year.

Figure 25 indicates annual grid sales against grid purchases of an unreliable grid-PV solar solution. The 385.7 kWh largest grid sales occurred in May while the least energy grid sales of about 85.7 kWh occurred in July. Although there were four dips in energy sales in the months of February (139.3 kWh), April (312.5 kWh), July (85.7 kWh), and November (289.3 kWh), the 85.7 kWh sold in July shows the worst sale figure for the year. The grid sales varied between 0.0 kW and 4.53 kW. The least grid

sales of 0.0 kW occurred around 24 March and 26 December, respectively. The majority of the grid sales were greater than 4.0 kW. The highest grid purchases of around 1.9 kW occurred on 24 July while the least grid purchases of 0.0 kW occurred around 31 May and 22 July, respectively. There were also clusters of grid purchases above 1.5 kW between 30 January and 2 March and 2 July to 14 July, and the average grid purchases hovered around 0.7 kW. The sale bifurcation point in July could have arisen because of much rainfall, cloud cover, or other contingencies like equipment or network failure. The energy purchase profile hovered between 100.0 kWh in February and 64.3 kWh in August. Although there was a gradual decline in energy purchase from February up to and including August, there was a steady rise, a plateau, around mid-September and mid-October, followed by an energy purchase decline up to the middle of November before an upward trajectory, thereafter.

Furthermore, the project cash flow comprises the net present costs (NPCs), annualized costs, and summary. The NPCs include US\$ 6354.00 capital, US\$ 936.23 operating costs, US\$ 2838.00 replacement costs, US\$ -290.29 salvage value (gain or profit from sales at the end of the plant's useful life), US\$ 0.00 resource costs, and US\$ 9838.00 total costs excluding salvage value. The annualized costs comprise US\$ 491.48 capital, US\$ 72.42 operating costs, US\$ 219.53 replacement costs, US\$ -22.44 salvage value (gain or profit), zero resource costs, and US\$ 760.99 total annualized costs. In summary, the total NPC was US\$ 9837.75, US\$ 6354.00 CAPEX, US\$ 269.51 OPEX, US\$ 0.1250 levelized COE/kWh, 12.1% return on invested capital (ROI), 16.0% inside rate of returns (IRR), simple payback period of 5.88 years, and zero (0.00 litre/year) fuel consumption costs.

5.2. Grid-Tied PV System. Figure 26 depicts the annual graph of grid sales against grid purchases for the PV grid-tied system. The maximum grid purchases of 1.3 kW occurred in January, March, April, and July. Also, grid purchases occurred throughout the year. Grid sales peaked around

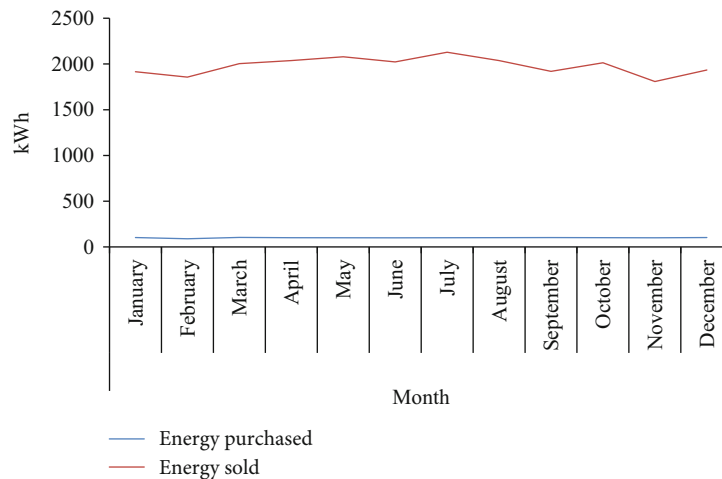


FIGURE 26: Grid sales against grid purchases for the grid-tied PV system.

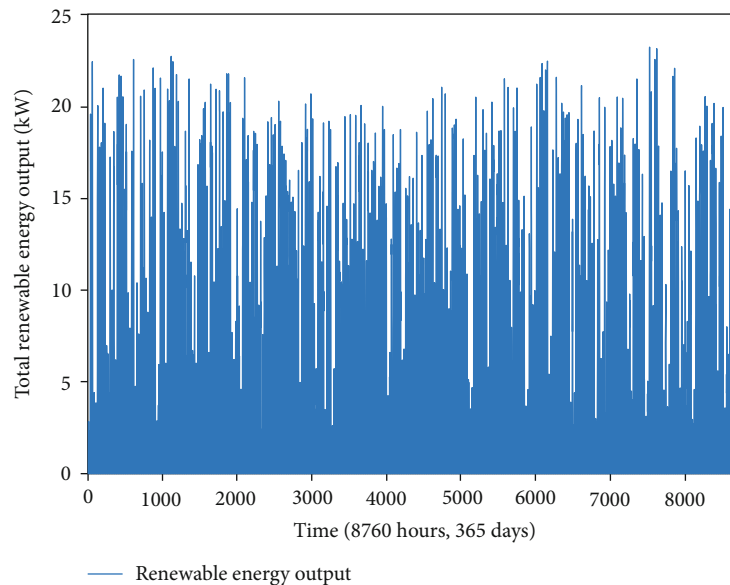


FIGURE 27: Total renewable output power of grid-PV system.

9.0 kW in May, July, August, October, and December, respectively. Also, the 2.2 kW least grid sales occurred around 18 May. Moreover, grid sales were close to 9.0 kW throughout the year. The total electrical load served varied between 0.0 kW and 9.1 kW. The highest energy sales of 2162.2 kWh occurred in July while the least energy sales of 1824.3 kWh occurred in November. Conversely, the energy purchase profile of 135.14 kWh was essentially the same throughout the year. This block pattern for the total electrical load served was experienced throughout the year. The excess electrical load production varied between 4.2 kW and 13.7 kW. The least electrical load served occurred in January while the largest electrical load served occurred in November. The highest total electrical load served was around 9.0 kW and the least electrical load served was 2.7 kW. The maximum total renew-

able power output was about 23.4 kW while the least total renewable power output was 2.7 kW.

The highest maximum total renewable power output was virtually the same throughout the year. The average total renewable power output hovered around 18.0 kW. The highest total electrical load served occurred around the first two weeks in November while the least total electrical load served occurred about 20 May. The largest global solar insolation of 1.2 kW/m² occurred in the first two weeks of November. The global solar insolation averaged around 0.8 kW/m² while the least global solar insolation of 0.13 kW/m² occurred around 11 February. Similarly, the largest PV solar power output of 23.4 kW occurred in the first two weeks of November while the least PV solar power output of 2.7 kW occurred around 6 February, 19 February, and 4 December, respectively. The

solar PV power output also averaged around 15.0 kW throughout the year.

Figure 27 shows the annual aggregate renewable output power. The highest aggregate renewable output power was 23.4 kW while the least aggregate renewable output power was 4.8 kW. By calculating the percentages, the maximum renewable penetration rate was about 100.0% while the least renewable penetration was 38.3%. The least renewable penetration rate occurred about 2 August while the highest penetration rate of 100.0% occurred on most days of the year. Similarly, the least total renewable power output occurred around 2 August while the highest occurred in the first two weeks of November. Additionally, the ratio between nominal renewable capacity and total nominal capacity and ratio between usable renewable capacity and total capacity were 100.0%, respectively. The ratio between total renewable production and load, ratio between total renewable production and generation, and one minus ratio between total nonrenewable production and load were 132.0%, 96.6%, and 100.0%, respectively.

The HOMER standard ratio between renewable output and load, ratio between renewable output and total generation, and one minus ratio between nonrenewable output and total load were 258.0%, 100.0%, and 100.0%, respectively. The gas emissions of carbon dioxide, SO_2 , and nitrogen oxides (NO_x) were 764.0 kg/year, 3.3 kg/year, and 1.62 kg/year, respectively. However, CO, unburned hydrocarbons, and particulate matter were zero kilogramme a year, respectively. The 22.0 kW was the rated PV system capacity; mean output power, 3.97 kW; and mean output energy, 95.4 kWh/day. The capacity factor was 18.1%, the total production was 34804.0 kWh/year, and the maximum output power was 23.2 kW. The PV penetration rate was 1324.0%; hours of operation, 4380 hours/year; and levelized cost, US\$ 0.0296/kWh.

Furthermore, the project cash flow consists of net present costs and annualized costs. The NPCs comprise US\$ 10876.00 capital, US\$ -22414.00 (gain or profit) operating costs, US\$ 630.05 replacement costs, US\$ -118.58 salvage value (gain), US\$ 0.00 resource costs, and US\$ -11026.00 (gain) total net present costs. The annualized costs consist of US\$ 841.30 capital, US\$ -1734.00 operating costs, US\$ 48.74 replacement costs, US\$ -9.17 salvage value (gain), US\$ 0.00 resource costs, and US\$ -852.93 total annualized costs (gain or profit). In sum, the total NPC costs were US\$ -11026.00 (gain), US\$ 10876.00 CAPEX, US\$ -1694.00, and US\$ -0.03232 levelized COE/kWh costs. The ROI was 18.6%; IRR, 22.7%; simple payback period, 4.35 years; and fuel consumption, 0.00 litre/year.

5.3. Grid-PV Minigrids

5.3.1. Grid-PV Minigrid without Storage. Figure 28 shows a sketch of the networked PV solar minigrid without storage. It consists of a grid supplying community AC load of 60385.6 kWh/year and a daily 20.46 kW peak power. The PV supplies the DC, which is connected to the converter for AC to DC or DC to AC conversion. The highest PV power output was 43.1 kW and occurred in the first two

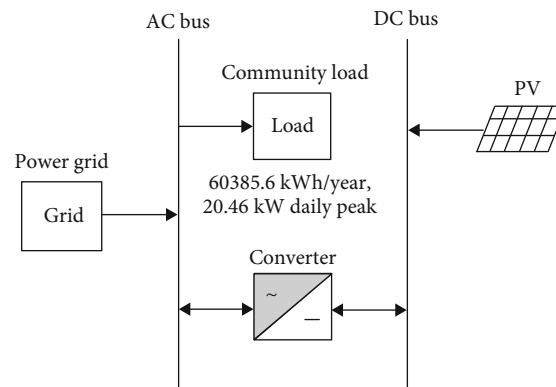


FIGURE 28: Schematic diagram of the hybrid grid-PV minigrid without storage.

weeks of November while the least power output about 6.4 kW occurred around 19 May. The average PV power output hovered around 38.0 kW. The maximum global solar insolation of 1.0 kW/m² occurred around 13 to 20 March while the least global solar insolation of 0.18 kW/m² occurred around 19 May. The largest total electrical load served peaked and occurred around 33.0 kW almost throughout the year while the least electrical energy production of 1.0 kW occurred around 30 April and 5 June, respectively.

Small grids have been used to generate electricity for rural and semiurban communes. They can be used as either temporary or permanent electricity supply solutions depending on the strategic importance of the locale or until when they could be connected to the public utility. Therefore, other policy, legal, financial, economic, and technological feasibilities would be considered before such extension could be implemented.

Figure 29 shows the annual graph of excess electricity production and total renewable power output of the grid-connected PV solar minigrid without storage. About 46.0 kW was the maximum total renewable output power which occurred within the first two weeks of November while the least total renewable power output was 5.3 kW and occurred around 20 May. Conversely, the highest excess electrical energy production of 10.7 kW occurred around the first two weeks in November while the least excess electricity production of 0.7 kW occurred about 5 June, respectively. Additionally, the ratio between nominal renewable capacity and total capacity and ratio between usable renewable capacity and total capacity were 100.0%, respectively. The ratio between total renewable production and load, ratio between the total amount of electrical energy produced annually by the renewable component and total system's annual generation capacity, and one minus ratio between total nonrenewable production and load were 72.6%, 67.1%, and 100.0%, respectively. The HOMER standard ratio between renewable output and load, ratio between renewable output and total generation, and one minus ratio between nonrenewable output and renewable output were 167.0%, 100.0%, and 100.0%, respectively.

Gas emissions for carbon dioxide, SO_2 , and nitrogen oxides (NO_x) were 20785.0 kg/year, 90.1 kg/year, and

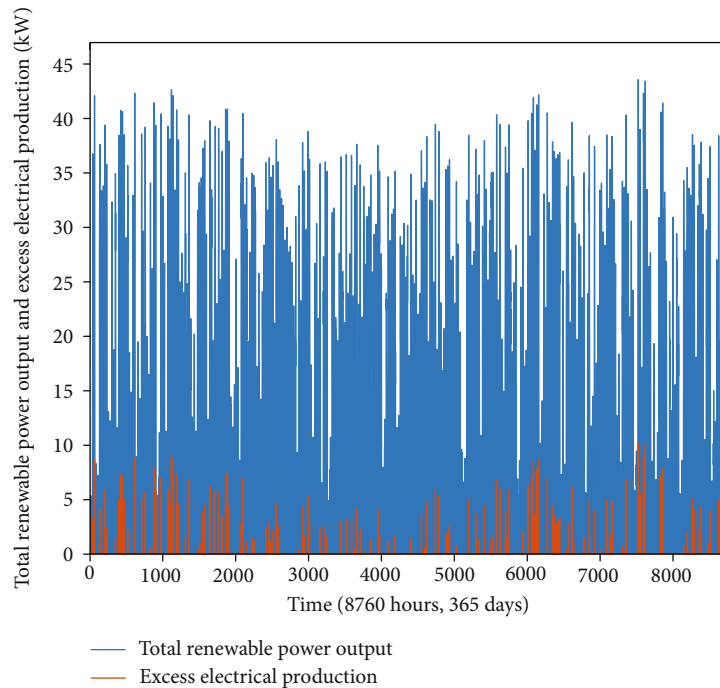


FIGURE 29: Aggregate renewable output power against excess electrical generation of the grid-PV small grid system without storage.

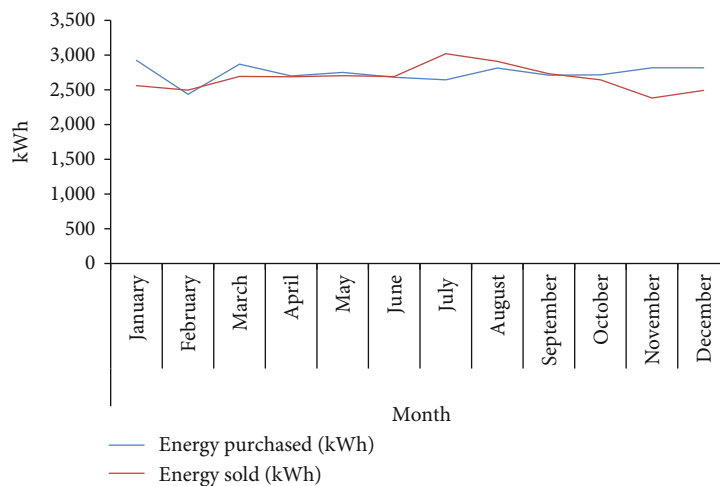


FIGURE 30: Grid sales and purchases for the grid-connected PV minigrid system without storage.

44.1 kg/year, respectively. Further, gas emissions for CO, unburned hydrocarbons, and particulate matter were zero kilogramme a year, respectively. Further, the PV system rated capacity was 45.3 kW; mean output power, 7.66 kW; mean output energy, 184.0 kWh/day; and capacity factor, 16.9%. The total energy production was 67087.0 kWh/year; maximum power output, 43.0 kW; PV penetration rate, 111.0%; hours of operation, 4380 hours/year; and levelized costs, US\$ 0.0971/kWh.

Figure 30 indicates the annual grid sales and purchases for hybrid grid-solar PV system without storage. The largest grid sales of 30.2 kW occurred around 7 January while the least grid sales of 0.4 kW occurred around 23 March, 14 May, 17 June, 2 August, 20–22 November, and 4 December,

respectively. The highest penetration rate was 132.6% and occurred in the first two weeks in November while the least was 74.5% and occurred around 10 February. However, the total renewable output power could not be ascertained because there was no visible purple coloration in the plateaued graph with spikes. If we assume that the plateau represents the mixture of total renewable power output, then 36.7 kW would be the largest total renewable output power while the least of 26.6 kW occurred around 11 February.

Moreover, the project cash flow comprises NPC and annualized costs. The NPCs include US\$ 80767.00 capital, US\$ 10605.00 operating costs, US\$ 2094.00 replacement costs, US\$ -1180.00 salvage value (gain or profit), and US\$ 0.00 resource costs, and the total cost is US\$ 92285.00 after

removing the salvage value. Additionally, the annualized costs comprise US\$ 6248.00 capital, US\$ 820.31 operating costs, US\$ 161.95 replacement costs, US\$ 91.27 -salvage value (gain or profit), and US\$ 0.00 resource costs, and the total annualized cost become US\$ 7139.00 after accounting for the residual value at the end of the plant's service life. In summary, the total net present cost is US\$ 92285.00; CAPEX, US\$ 80767.00; OPEX, US\$ 890.99; levelized COE/kWh cost, US\$ 0.0742; IRR, 16.5%; simple payback period, 5.91 years; and fuel consumption cost, 0.0 litre/year.

5.3.2. Grid-Connected PV Minigrad Systems with Storage. Figure 31 depicts the schematic diagram for the PV solar minigrad with storage. The system comprises a grid that supplies AC electricity to the community while the solar PV system supplies DC. The community load is 60385.6 kWh/year and has a daily 20.46 kW peak. The converter charges the AC load to DC for storage and changes back to AC if needed while the PV solar can be converted to AC from DC when the demand arises.

The grid-tied photovoltaic systems work with the regional utility grid such that excess solar electricity generation with respect to the load is fed to the grid and any shortfall from the solar PV systems is supplied by the grid. The islanded solar PV systems supply electrical loads and charge batteries during sunlight hours but can only supply loads from storage batteries through converters when the sun is not accessible.

Equipment that makes up hybrid grid-solar PV systems includes inverters, PV arrays, (with and without) storage batteries, electricity meters, AC breakers, fuses, isolators, safety switches, earthing, and cabling. The electricity generated from PV arrays flows in one direction only and is converted to AC by inverters to suitable voltages and frequencies. The prime parameters considered in choosing converters depend on their highest and lowest voltages and electricity conversion efficiencies. Further, hybrid grid-solar PV systems are beneficial because, when excess electricity is produced by the PV systems, it is fed into the grid. This relieves the grid of more carbon footprint and greenhouse gas emissions and avoided production costs.

For either maintenance or testing purposes, the PV system is disconnected from the converter. Thus, isolator switches of maximum rating, inverter safety switches, and adequately sized and precisely rated cables are required for workable systems.

Further, hybrid PV-grid systems without batteries are easy to use and less expensive to implement. Although this system seems reliable, if there is grid failure without battery storage, critical sectors and services could be negatively affected. Therefore, hybrid PV-grid systems with battery storage offer uninterruptible supply, but this benefit comes at a cost of higher financial expenses and technological complexities.

Figure 32 indicates the annual grid purchases and sales for the PV solar minigrad with storage. The highest energy was 3305.6 kWh in July while 1968.8 kWh was the least energy sold that occurred in November. Similarly, the largest energy purchased was about 2916.7 kWh and occurred

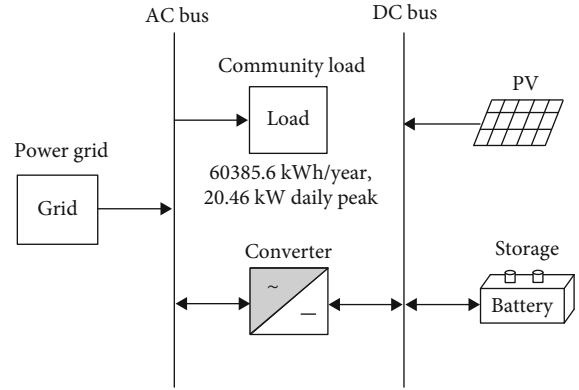


FIGURE 31: Schematic diagram for the PV grid-connected minigrad system with storage.

in March while the least energy purchased of about 2333.3 kWh occurred in February. The largest total electrical load served was around 36.3 kW for the generality of the sampled data while the smallest was 0.0 kW and occurred about 20 May. Conversely, the largest excess electricity production of 11.1 kW occurred around the first two weeks of November while the least of 0.0 kW occurred around 20 May.

Renewable energy penetration percentage (P_{ren}) is the ratio between total renewable electrical power output (kW) and total electrical load (L_{served}) served (kW). Also, E_{unmet} is the total unmet load (kWh/year), E_{demand} is the aggregate annual electrical demand (primary and deferred) (kWh/year), $E_{served,prim,AC}$ is the AC primary load served (kWh/year), $E_{served,prim,DC}$ is the DC primary load served (kWh/year), $E_{served,def}$ is the deferred load served (kWh/year), and $E_{grid,sales}$ is the energy sold to the grid (kWh/year) [23, 36].

$$P_{ren} \% = \frac{P_{ren}}{L_{served}},$$

$$f_{unmet} = \frac{E_{unmet}}{E_{demand}},$$

$$f_{ren} = \frac{1 - E_{nonren} + H_{non}}{E_{served} + H_{served}},$$

$$E_{served} = E_{served,Acprim} + E_{served,DCprim} + E_{served,def} + E_{grid,sales}, \quad (32)$$

where f_{unmet} is the unmet demand factor (%), E_{unmet} is the unmet load (kWh/year), E_{demand} is the load demand (kWh/year), f_{ren} is the renewable load factor (%), E_{nonren} is the nonrenewable load (kWh/year), H_{nonren} is the nonrenewable enthalpy (kWh/year), E_{served} is the energy served (kWh/year), and H_{served} is the served enthalpy (kWh/year).

Figure 33 depicts the annual renewable penetration rate for the PV minigrad with storage. The highest renewable penetration rate was 10.71% and peaked around 106.1% for the majority of the periods throughout the year while the least

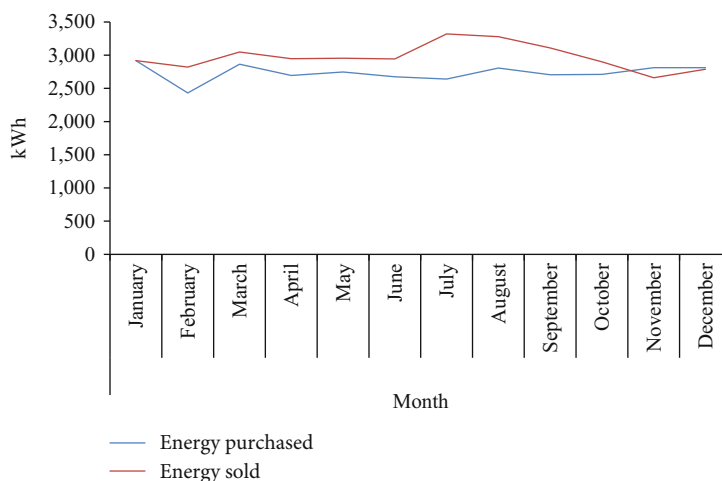


FIGURE 32: Grid sales against purchases of the hybrid PV-grid small-grid system with storage.

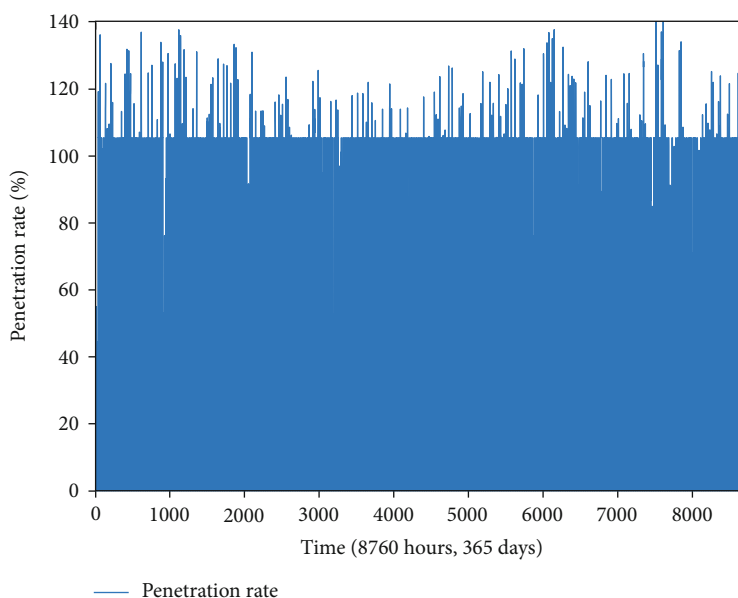


FIGURE 33: Renewable penetration and total renewable power output for the PV grid-connected minigrid system with storage.

of about 25.3% occurred around 8 January. Similarly, the largest total renewable power output of about 44.2 kW occurred within the first two weeks of November while the least of about 7.9 kW occurred around 8 January, respectively. In brief, the total NPC cost was US\$ 92799.84, US\$ 80311 CAPEX, US\$ 966.05 OPEX, US\$ 0.0747 LCOE, and 12.1% return on invested capital. Further, IRR was 16.6%, simple payback period was 5.86 years, and fuel consumption was 0 l/year. The project cash flow comprises NPC and annualized costs. Consequently, the capital was US\$ 80311; operating cost, US\$ 10411; replacement cost, US\$ 2891; salvage value (gain or profit), US\$ -544.21; resource cost, US\$ 0; and total net present cost, US\$ 92800. The annualized costs comprise US\$ 6121 capital, US\$784.47 operating cost, US\$ 223.67 replacement cost, US\$ -42.1 salvage value, US\$ 0

resource cost, and US\$7178 total annualized cost (less salvage value). Further, the ratio between nominal renewable capacity and total nominal capacity and ratio between usable renewable capacity and total capacity were 100%, respectively. Also, the ratio between total renewable production and load, ratio between total renewable production and generation, and one minus ratio between total nonrenewable production and load were 70.8%, 67.4%, and 100%, respectively.

In addition, HOMER standards for the ratio between renewable output and load, ratio between renewable output and total generation, and one minus ratio between nonrenewable output and total load were 141%, 100%, and 100%, respectively. The gas emissions for carbon dioxide, SO₂, and nitrogen oxides (NO_x) were 200738.0 kg/year, 89.9 kg/year, and 44.0 kg/year, respectively. Also, gas

emissions for CO, unburned hydrocarbons, and particulate matter were zero kilogramme a year, respectively. Nevertheless, the grid-connected PV minigrid with storage was rated 45.9kW and the mean output power was 7.76kW while the mean output energy was 186kWh/day. The capacity factor was 16.9%, the total production was 67993kWh/year, and the maximum output power was 43.6kW. The PV penetration rate was 113%; operation hours a year, 4380; and LCOE, US\$ 0.0904/kWh. Without net metering, the HOMER software determines aggregate annual energy charge from

$$C_{\text{grid,energy}} = \sum_i^{\text{rates}} \sum_j^{12} E_{\text{gridpurchase},i,j} \times C_{\text{power},i} - \sum_i^{\text{rates}} \sum_j^{12} E_{\text{gridsales},i,j} \times C_{\text{sellback},i} \quad (33)$$

With net monthly metering and monthly net generation determined, the aggregate annual energy charge becomes

$$C_{\text{grid,energy}} = \sum_i^{\text{rates}} \sum_j^{12} \begin{cases} E_{\text{netgridpurchases},i,j} \times C_{\text{power},i}, & \text{if } E_{\text{netgridpurchases},i,j} \geq 0, \\ E_{\text{netgridpurchases},i,j} \times C_{\text{sellback},i}, & \text{if } E_{\text{netgridpurchases},i,j} < 0. \end{cases} \quad (34)$$

With net metering and annual net generation determined, the aggregate energy charge becomes

$$C_{\text{grid,energy}} = \sum_i^{\text{rates}} \begin{cases} E_{\text{netgridpurchases},i} \times C_{\text{power},i}, & \text{if } E_{\text{netgridpurchases},i} \geq 0, \\ E_{\text{netgridpurchases},i} \times C_{\text{sellback}}, & \text{if } E_{\text{netgridpurchases},i} < 0. \end{cases} \quad (35)$$

The total annual grid demand charge can be expressed as

$$C_{\text{grid,demand}} = \sum_i^{\text{rates}} \sum_j^{12} P_{\text{grid,peak},i,j} \times C_{\text{demand},i} \quad (36)$$

where $P_{\text{grid,peak},i,j}$ is the hourly grid peak demand in month j during time rate i , $C_{\text{demand},i}$ is the grid demand measure at rate i (US\$/kW/month), and $C_{\text{sellback},i}$ is the sellback measure at rate i (US\$/kWh). Also, $C_{\text{power},i}$ is the grid power price at rate i (US\$/kWh), $E_{\text{netgridpurchases},i}$ is the annual net grid purchases at time rate i (kWh), $E_{\text{netgridpurchases},i,j}$ is the monthly net grid purchases in month j at time rate i (kWh), and $E_{\text{gridpurchases},i,j}$ is the energy quantity purchased from the grid in month j at time rate i (kWh) [23, 36].

5.4. Summary. Section 5 contains the modeling and optimization of hybrid PV-grid systems. In this section, an individual household load connected to an unreliable grid and the hybrid PV-grid system to provide reliable electricity power supply to an individual household and community load connected to two hybrid PV-grid systems with or without storage were considered. The household connected to the unreliable grid produced excess electricity of 6999.0 kWh/year (51.7%), unmet electric load of 1.24 kWh/year (0.0471%), and capacity shortage of 2.05 kWh/year (0.0781%). Further, the

renewable energy fraction was 84.2%; maximum renewable energy penetration, 4645.0%; CO₂ emissions, 609.0 kg/year; nitrogen oxide (NO_x) emissions, 1.29 kg/year; and LCOE, US\$ 0.1250/kWh. For the grid-tied PV system, excess electricity was 8301 kWh/year (23.1%); unmet electric load, 0.0 kWh/year; capacity shortage, 0.0 kWh/year; renewable energy fraction, 95.4%; and maximum renewable energy penetration, 258.0%. Also, CO₂, SO₂, and nitrogen oxide (NO_x) gas emissions were 764.0 kg/year, 3.31 kg/year, and 1.62 kg/year, respectively, and LCOE was US\$ -0.03232/kWh. The negative sign on LCOE indicates that the grid-tied PV system owner will not need to pay any US\$ amount per electricity used but the utility grid will start paying US\$ 0.0323/kWh to the owner of the grid-tied PV system after its complete installation. For the grid-connected PV minigrid without storage, the excess electricity produced was 983.0 kWh/year (0.999%); unsatisfied electric load, 0.0 kWh; and annual capacity shortage, 0.0 kWh. Also, the renewable fraction was 65.6%; the maximum renewable energy penetration was 133.0%; CO₂ and nitrogen oxide (NO_x) gas emissions were 20896.0 kg/year and 44.3 kg/year, respectively; and LCOE was US\$ 0.07420/kWh. For the grid-connected PV minigrid with storage, the excess electricity was 1403 kWh/year (1.39%); the unsatisfied electrical load was 0.0 kWh; the shortage capacity was 0.0 kWh; the renewable fraction was 65.8%; the maximum renewable penetration

was 141.0%; CO₂ and nitrogen oxide (NO_x) gas emissions were 20738.0 kg/year and 44.0 kg/year, respectively; and LCOE was US\$ 0.07472/kWh.

6. Conclusion

The geographical location of Rwanda and incentives for renewable energy with other facts such as investment procedures and favorable conditions such as doing business conditions and living conditions (improved roofing materials, increased number of dwellings constructed close to each other (Umudugu), and households with mobile phones and internet connections which can facilitate in spreading or communicating to people about the new energy technology in Rwanda) can contribute to the maximum and efficient exploitation of solar energy in Rwanda. This is so because this resource potential has not been fully exploited. Further, the analysis made on the monthly and annual energy fed into the national grid by the 8.5 MW Rwamagana solar power plant reveals good hope and success for anyone who would think to invest in solar energy technologies in Rwanda.

For off-grid users who have no access to electricity, photovoltaic solar technologies were modeled via the stand-alone solar system and solar minigrid technology solutions. The model of a stand-alone photovoltaic system for a 7.204 kWh/day household load located in Rutsiro, Rwanda (1°56.3'S, 29°19.5'E) reveals that the system was annually able to produce excess electricity of 6445 kWh (67.5%), with unmet electric load of 0.649 kWh/year (0.0247%) and capacity shortage of 2.54 kWh (0.0965%). The system completely used renewable energy technology (100% renewable fraction) with 4129% maximum renewable penetration, US\$ 0.6155 LCOE (levelized cost of electricity), and zero hydrocarbon and greenhouse gas emissions. For the community load of 165.44 kWh/day, two solar minigrid technologies were simulated and optimized. The off-grid photovoltaic solar system with storage presented the excess electricity of 51348 kWh/year (42.6%), unmet electric load of 42.9 kWh/year (0.0711%), and capacity shortage of 56.4 kWh/year (0.0934%) with renewable fraction of 100% and maximum renewable energy penetration of 2663%. The system presented zero carbon and gas emissions, and its LCOE is US\$ 0.3903. Another minigrid solar technology composed of photovoltaic solar and hydropower turbine, together with its storage, presented the excess electricity of 319791 kWh/year (83.9%), unmet electric load of 38.6 kWh (0.064%), capacity shortage of 60.3 kWh/year (0.999%), renewable fraction of 100% and maximum renewable penetration of 9683.0%, LCOE of US\$ 0.7239/kWh, and zero hydrocarbon and greenhouse gas emissions.

For on-grid users who access electricity from an unreliable grid characterized by frequent blackouts, power outages, and high electricity tariffs, photovoltaic solution technologies were modeled and optimized via a photovoltaic solar system connected through ATS (automatic transfer switch), grid-tied photovoltaic systems, or grid-connected photovoltaic solar minigrids. The photovoltaic solar system with storage connected via an ATS reveals that with an AC primary load,

it was able to produce excess electricity of 6999 kWh/year (51.7%), unmet electric load of 1.24 kWh/year (0.0471%), capacity shortage of 2.05 kWh/year (0.0781%), renewable energy fraction of 84.2%, maximum renewable energy penetration of 4645.0%, CO₂ emissions of 609.0 kg/year, nitrogen oxide (NO_x) gas emissions of 1.29 kg/year, and LCOE of US\$ 0.1250/kWh. For the grid-tied photovoltaic solar system, the excess electricity was 8301.0 kWh/year (23.1%); the unsatisfied electric load was 0.0 kWh/year; the shortage capacity was 0.0 kWh/year; the renewable energy fraction was 95.4%; the maximum renewable energy penetration was 258.0%; CO₂, SO₂, and nitrogen oxide (NO_x) gas emissions were 764.0 kg/year, 3.32 kg/year, and 1.62 kg/year, respectively; and LCOE was US\$ -0.03232. For the grid-connected photovoltaic solar minigrid without storage, the excess electricity produced was 983.0 kWh (0.999%); the unsatisfied electric load was 0.0 kWh; the capacity shortage was 0.0 kWh/year; the renewable fraction was 65.6%; the maximum renewable energy penetration was 133.0%; CO₂ and nitrogen oxide (NO_x) gas emissions were 20896.0 kg/year and 44.3 kg/year, respectively; and LCOE was US\$ 0.07420. For the grid-connected photovoltaic solar minigrid with storage, the excess electricity was 1403.0 kWh (1.39%); the unsatisfied electric load was 0.0 kWh; the shortage capacity was 0.0 kWh; the renewable fraction was 65.8%; the maximum renewable penetration was 141.0%; CO₂ and nitrogen oxide (NO_x) gas emissions were 20738.0 kg/year and 44.0 kg/year; and LCOE was US\$ 0.07472.

Referring to the computational simulation and optimization results from the HOMER software, supplying electricity through solar-operated minigrid technologies would increase electricity access to the population. For the off-grid users, solar photovoltaic technology with storage was recommended in this paper. And for the on-grid users, either of the two hybrid grid-PV system technologies with or without storage is recommended because their LCOEs were below the normal tariffs of electricity in Rwanda.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interests.

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